To Revise or not to Revise? Evidence Sufficiency and Uptake in the Residual Stream of Retrieval-Augmented LLMs

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Abstract

Since the wide adoption of Large Language Models (LLMs), Retrieval Augmented Generation (RAG) has thrived in the specialised literature as a way to ground LLMs' responses on custom knowledge bases. In this context, a central question about the internal mechanics of the model remains unanswered: when new, retrieved evidence becomes sufficient to answer a question, does an LLM undergo an internal state transition that predicts whether it will use that evidence, before it generates a token? Here, we directly investigate this question. To do so, we develop a framework consisting of paired forward passes over semantically defined *evidence trajectories* for multi-hop questions: for each question we present nested contexts that add required paragraphs to answer the question, paired with various forms of distractors. By fitting linear probes on the paired activation update from the internal residual streams of the model, we then show that information about *sufficiency* (the increment that correctly completes the evidence) is recoverable at around 0.95 AUROC across three model families. We then show a second, orthogonal direction predicting *uptake*, i.e. whether decisive evidence will actually be integrated rather than ignored, at 0.78 AUROC. Causal tests highlight how the found directions correlate with but do not cause a model behaviour, while a causal direction acting as an abstention gate (whether the model should answer or wait for more evidence) can be found via the mean-differences of sufficient versus insufficient states. Finally, we replicate every finding on HotpotQA, on RAGBench's naturally retrieved documents across five domains, and on a synthetic benchmark with paraphrase, partial and contradictory evidence, showing that directions learned on Wikipedia do transfer, but only partially to other domains.

1 Introduction

Retrieval-Augmented Generation (RAG) is a recent development combining traditional Information Retrieval and recent advances in Natural Language Processing, where a Large Language Model (LLM) is supplied with retrieved passages and asked to find the relevant information in such passages in order to answer a knowledge-grounded question (Lewis et al., 2020). Since the whole point of RAG systems is to optimize the information available to the LLM by retrieving the most relevant one(s), a stress test for these systems is the one concerning multi-hop questions, where the answer depends on multiple, incremental and separate pieces of evidence (Trivedi et al., 2022). In such context, two critical, under-studied elements are the internal ability of the LLM to assess that the supplied evidence is now *sufficient*, and to judge the successful *uptake* of the decisive passage into its answer computation. Prior work has shown that residual-stream activations carry signals about context-versus-parametric conflict and context reliance (Zhao et al., 2024, 2025), but these are typically identified on single contexts, and they typically do not analyse if and when the model's internal dynamics can predict the two above concepts in the absence of relevant parametric knowledge to answer the question.

We then use *increments*, i.e. incremental additions of context, to investigate these elements. For each multi-hop question (i.e. a question requiring several pieces of information at different steps of the answer), we construct nested evidence contexts $C_0 \subset C_1 \subset \cdots$ and run a fresh, stateless forward pass on each, reading the residual stream at a fixed anchor before the answer generation. The difference between the updated and previous contexts $\Delta h_k = h(C_k) - h(C_{k-1})$ isolates what the added evidence context C_k did to the model's state. We use such difference to build positive and negative examples to train linear probes for predicting *suffi-*

ciency (whether the model has gained all the relevant information to answer the question), deriving the positive examples by the *decisive increments* (i.e. the question with the added relevant paragraph) and the negative ones from both the *non-sufficient increments* (added information is relevant but still not enough), as well as matched *distractor increments* where we substitute the correct evidence context with irrelevant ones, so as to prevent the probes overfitting on nuisances such as context length. We similarly derive training samples and fit linear probes for the other binary task of *uptake* (whether the model will answer correctly, given it has gained all the relevant information), but also for *correction* (answer will go from incorrect to correct, given the update), *stability* (answer won't change) and *revision* (answer will change) so as to separately analyse the various information recoverable from the model internal states. From the experiments, we find that sufficiency and uptake, as well as the other additional information, are all indeed recoverable from the internal states, but steering the model along the identified dimensions from each linear probe does not alter significantly models' behaviour, indicating that such dimensions are readouts (i.e. correlates) rather than causal mechanisms for the relative task. A further text-only baseline shows that sufficiency is recoverable from text alone, while uptake is not. We further employ activation addition (Turner et al., 2024) to find a true causal direction for sufficiency which can be obtained via the mean-difference between sufficient and insufficient states and, we show, acts as a gate for the model to abstain from answering, complementing previous literature on LLM abstention when steering *confidence* directions (Kumaran et al., 2026). Finally, we show that our results hold for two additional model families and on external benchmarks, both by re-training and by transferring the learned probes.

Our contributions are as follows.

- **A matched-increment protocol** for reading evidence-sufficiency and evidence-uptake states from single forward passes before answer generation.
- We show that **sufficiency and uptake are linearly decodable** from residual activation of the model and that they have nearly orthogonal directions. We further show that the found directions are readouts and not causal mechanisms.
- We find a **causally active direction** (the mean difference between sufficient and insufficient states) which acts as a steerable abstention gate in the tested models' families.
- We show the **generalisability** of our method by replicating results on HotpotQA (Yang et al., 2018), on RAGBench's (Friel et al., 2024) naturally retrieved documents across five domains and single-hop questions, and on a synthetic benchmark with paraphrase, partial and contradictory evidence. In this context, we build our own benchmark, the Evidence State Transition Benchmark (ESTB), which aggregates these sources under one labelling scheme and adds retrieved and rewritten evidence; we release it to foster research in multi-hop reasoning and incremental evidence processing.

The picture that emerges is that evidence sufficiency and uptake are cleanly *readable* before generation, that the readout of uptake is not recoverable from text, and that neither readout is a *lever*: the linear probe finds a correlate of the state, while the mechanism that turns evidence into an answer is high-dimensional and content-bearing.

2 Related work

3 Experimental setup

Data. MuSiQue (Trivedi et al., 2022) provides 2–4-hop questions with gold supporting paragraphs, per-hop decompositions with sub-answers and relation types, and distractors; our main runs use its two-hop training questions, a second run 3–4-hop questions, and 2WikiMultiHopQA (Ho et al., 2020) is the transfer set. For each model we screen a pool of 4,000 questions closed-book and keep 2,000 that the model answers incorrectly without evidence (closed-book accuracy is 1.4% for Qwen, 4.0% for Llama, 1.3% for 3–4-hop questions), so every increment concerns evidence the model actually needs. We keep the whole trajectory for every question, including stages after the answer first becomes correct, to avoid conditioning the data on the model's behaviour.

Models. Qwen2.5-7B-Instruct (28 layers, $d = 3584$), Llama-3.1-8B-Instruct (32 layers, $d = 4096$) and Mistral-7B-Instruct-v0.3 (32 layers, $d = 4096$), all in bfloat16 with greedy decoding of at most 16 tokens. Prompts use each model's chat template.

Prompt and anchor. Every stateless stage uses the same template: an instruction ("Use only the supplied evidence to answer the question. If the evidence is insufficient, output INSUFFICIENT."), the evidence block, the question and an answer cue. We read the residual stream after every decoder block at the last prompt token, the position that emits the first answer token, so all readouts are pre-generation. An *evidence-augmented* regime ("use the evidence together with your own knowledge") is run in parallel and gives the same results (Section F); we report the evidence-only regime unless stated.

Evidence conditions. For a question with gold hop paragraphs $g_1, \ldots, g_n$ we build (Section A): no evidence; matched distractors only; prefix chains in the canonical, reverse and (for $n \geq 3$) random hop orders, the last prefix being *minimally sufficient*; leave-one-out contexts; the sufficient context plus a distractor, a duplicate support or a conflicting falsified paragraph; and the full 20-paragraph context. New paragraphs are inserted at a random position, so increments are textually minimal while the decisive paragraph's position is randomised. Falsified paragraphs swap the hop's sub-answer for another question's sub-answer of the same relation type; answer-string controls append a "not to be confused with {answer}" hatnote to a distractor.

Increments and labels. From every penultimate state S (all hops but one) we form the decisive increment $S \rightarrow S{+}g_{\text{last}}$ and four matched controls from the same S: distractor, redundant, false and answer-string. With first-hop, post-sufficiency and full-context increments this yields 17 conditions and 17 increments per two-hop question (35,008 conditions per model). Each increment carries $y^{\text{suff}} = \mathbf{1}[C_k \text{ sufficient}, C_{k-1} \text{ not}]$ and the model's behavioural transition, defining *uptake* (decisive increments: answer becomes correct or not), *correction* (wrong or abstaining $\rightarrow$ correct or not), *stability* (correct $\rightarrow$ correct or not) and *revision* (answer changed). Correctness is exact match on any alias or containment within a short prediction.

Behaviour and splits. All three models abstain on 94–97% of distractor-only contexts and answer 64–75% of minimally sufficient ones with 36–47% correct; falsified final hops halve abstention (Table 6, Section C). Probes are trained on 70% of questions and evaluated on the rest under four groupings never shared between train and test: question id, gold document (connected components of questions sharing a supporting page), answer entity, and relation-type tuple. We report the document split by default and all four in Table 10 (Section E).

4 Method

Paired updates and probes. Let $h^{(l)}(C) \in \mathbb{R}^d$ be the residual stream after block l at the anchor for context C. For increment $C_{k-1} \rightarrow C_k$ we form $\Delta h^{(l)} = h^{(l)}(C_k) - h^{(l)}(C_{k-1})$ and, for each task and layer, fit an ℓ_2-regularised logistic regression on standardised features with class balancing, selecting the layer by grouped three-fold cross-validation on training questions only. We also report the *difference-of-means* direction $v_{\text{dm}} \propto \mathbb{E}[\Delta h \mid y{=}1] - \mathbb{E}[\Delta h \mid y{=}0]$. Directions are unit vectors in raw activation space.

State probes. Update probes classify *transitions*. A second family classifies single *contexts*. A *state probe* is fitted with the same procedure on the raw anchor activation $h(C)$ of one forward pass, one example per condition rather than per increment. The *state-sufficiency* probe labels a condition 1 if it contains every gold hop: the minimal sufficient context, the sufficient context plus a distractor or a duplicate, and the full 20-paragraph context. It labels it 0 otherwise: no evidence, distractors only, one hop, and the penultimate state plus a distractor, duplicate or answer-string control. Conditions containing a falsified paragraph are excluded because their sufficiency is ill-defined. This gives about 20,000 training and 8,800 test conditions per model, with 31% positives. The *state-correctness* probe labels every condition by whether the model's answer is correct, and the *abstention-flag* probe by whether the model outputs INSUFFICIENT. The two families answer different questions. The state probe asks whether a context, taken on its own, contains all the evidence. The update probe asks whether *this* paragraph completed it: its negatives include increments that end in a sufficient context, such as a distractor added after sufficiency, and controls that start from the same penultimate state, which the state probe never has to separate. Their AUROCs are therefore not comparable. Section 5.1 measures how much of the update readout is already present in either endpoint state, and whether either readout is just the model's planned output.

Nuisance analysis. A paired difference also encodes added tokens, context length, answer-string presence, passage relevance and the change in the output distribution. We build a nuisance vector z per increment (token counts, gold-answer log-probability change, next-token entropy and top-probability change, answer-string presence, TF-IDF relevance, insertion position, and a 16-dimensional TF-IDF-SVD embedding of the added text; Section B) and report (i) a nuisance-only classifier, (ii) the probe on residuals of a ridge regression of Δh on z, and (iii) the AUROC gain of adding out-of-fold probe scores to the nuisance model.

Text baselines. Text-only classifiers on the same increments and splits: TF-IDF logistic regression and a fine-tuned DeBERTa-v3-base cross-encoder, on "question [SEP] added paragraph" alone and with the previous context appended.

Transfer. Learned directions are applied without retraining to (i) the *conversational revision* protocol, in which the model sees its own previous answer A_k and only the new evidence and is told to change its answer only if the total evidence warrants it, pairing $h_{\text{conv}}(C_k \to C_{k+1}) - h(C_k)$; (ii) 2WikiMultiHopQA; (iii) 3–4-hop MuSiQue; and (iv) the datasets of Section 7. Model-family transfer is replication of the procedure, since hidden sizes differ.

Causal tests. Interventions act on held-out questions of the document split. *Steering* adds $\alpha\, u\, v$ from the anchor onward at layer l (or all positions), where u is the separation between the sufficient and insufficient state means along v on training data, so $\alpha = 1$ moves an activation by one sufficient-minus-insufficient gap; a random unit direction with the same ℓ_2 magnitude is the control. *Projection-out* sets the component along v to its insufficient-state mean at all layers from l onward. *Patching* runs the insufficient prompt C_{n-1} with its anchor residual replaced, at every layer from l onward, by h_{suff} (full), $h_{\text{ins}} + \text{proj}_v(\Delta)$ (parallel), $h_{\text{ins}} + \Delta - \text{proj}_v(\Delta)$ (orthogonal), the projection of Δ onto a random direction, onto the top-k principal components of training deltas, or onto the gold answer's first-token unembedding direction. Outcomes are the rate of non-abstention ("answered") and of correct answers on 150 held-out questions per set; for false-evidence contexts we also report the rate of producing the injected false answer.

5 Readout results

This section asks whether sufficiency, uptake and the other behavioural transitions can be read linearly from the residual stream before the model generates its answer. Every number is the AUROC of a logistic probe on held-out questions, at the layer chosen by cross-validation on the training questions. Unless stated otherwise we use the *document split*, in which no Wikipedia page that supports a test question also supports a training question, so a probe cannot succeed by memorising facts about specific pages.

Sufficiency is linearly readable and robust. A probe trained to recognise the decisive increment, the one that adds the last missing paragraph, separates it from all other increments at 0.948–0.952 AUROC in the three models (Table 1). The signal is strongest at 50–60% of the network's depth (Fig. 1). The harder test compares the decisive increment only with its four *matched controls*, which start from the same context with one paragraph missing and add, instead of that paragraph, a length-matched distractor, a duplicate of a paragraph already present, a falsified copy of the missing paragraph, or an irrelevant paragraph that mentions the answer string. Against these controls the probe still reaches 0.91–0.92. Surface properties do not explain the result either. A classifier that sees only the nuisance covariates reaches 0.86–0.87, adding the probe score to it gains 0.11–0.12, and a probe trained after the covariates have been regressed out of the activations still reaches 0.94–0.96. The result does not depend on how questions are split between training and test: splits that keep questions, supporting documents, answer entities or relation types disjoint all give 0.94–0.96 (Table 10). The state-sufficiency probe, which classifies whole contexts rather than transitions, reaches 0.99, but part of that comes from paragraph count; Section 5.1 separates the two.

Figure 2 projects held-out increments onto the sufficiency direction, that is, it computes their score along the probe's weight vector. The kinds of increment are ordered as one would expect if the direction encoded "the evidence has just become complete": decisive $>$ false $>$ distractor $>$ duplicate $>$ answer-string control $>$ a gold paragraph that does not yet complete the evidence. Increments added after the context is already sufficient, and the first hop, score far below. The falsified paragraph scoring second indicates that the direction

	Qwen2.5-7B			Llama-3.1-8B			Mistral-7B		
Task	AUROC [95% CI]	nuis.	gain	AUROC [95% CI]	nuis.	gain	AUROC [95% CI]	nuis.	gain
Sufficiency	0.948 [0.94,0.95]	0.87	0.11	0.952 [0.95,0.96]	0.86	0.12	0.949 [0.94,0.95]	0.87	0.11
Sufficiency (matched)	0.911 [0.90,0.92]	0.90	0.07	0.918 [0.91,0.93]	0.89	0.08	0.910 [0.90,0.92]	0.90	0.07
Uptake	0.775 [0.74,0.80]	0.66	0.14	0.783 [0.75,0.81]	0.58	0.21	0.771 [0.74,0.80]	0.59	0.18
Correction	0.730 [0.70,0.76]	0.81	0.01	0.817 [0.79,0.84]	0.83	0.06	0.792 [0.77,0.81]	0.85	0.03
Stability	0.750 [0.72,0.78]	0.92	−0.01	0.869 [0.85,0.89]	0.90	0.02	0.800 [0.77,0.83]	0.90	0.01
State sufficiency	0.992 [0.99,0.99]	–	–	0.994 [0.99,0.99]	–	–	0.996 [1.00,1.00]	–	–

Table 1: Held-out AUROC of the best-layer probe on paired activation updates Δh (document split; 95% question-level bootstrap intervals). "nuis.": AUROC of a classifier that sees only surface properties of the increment (nuisance covariates such as length change, presence of the answer string, lexical relevance and change in output confidence). "gain": AUROC added when the probe score is given to that classifier. The last row is a different task: the state-sufficiency probe classifies single contexts (raw activation $h(C)$, sufficient vs. insufficient), not transitions, so its AUROC is not comparable with the rows above (Sections 4 and 5.1).

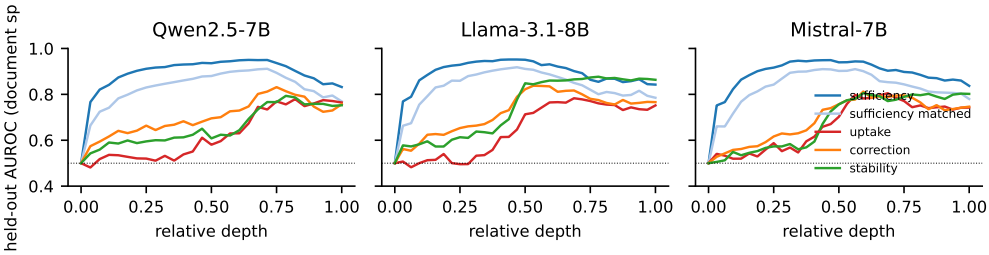


Figure 1: Held-out AUROC by relative depth for the update tasks (document split). Sufficiency peaks around mid-depth in every model; uptake and correction emerge later.

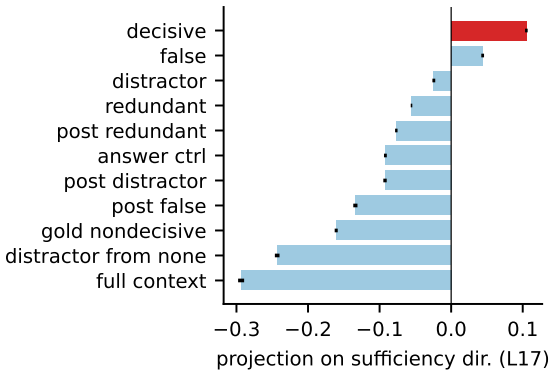


Figure 2: Mean projection of held-out increments on the sufficiency direction, by kind of increment (Qwen2.5-7B, document split; standard errors). Red: decisive.

encodes "the final slot has been filled" rather than truth, which is all the model can register before it generates.

Uptake is a separate, later signal. Among decisive increments, a probe can also predict whether the model will use the new paragraph, that is, whether its answer becomes correct, or will ignore it. This uptake probe reaches 0.77–0.78 AUROC on the document split and 0.85 on the question split, at layers near the end of the network. Uptake is also the task where the activations add most beyond surface properties (+0.14 to +0.21 over the nuisance classifier). It is a different signal from sufficiency. The two directions are nearly orthogonal (cosine 0.02–0.03), and the sufficiency direction scores only 0.54–0.60 when used to predict uptake. Uptake does share a direction with *correction*, the transition from a wrong or abstaining answer to a correct one (cosine 0.30–0.34; each direction predicts the other task at 0.82 AUROC; Fig. 5, Section G). Correction and stability, by contrast, are largely predictable from the change in the model's output distribution. Nuisance-only classifiers, which include that change, reach 0.81–0.85 on correction and 0.90–0.92 on stability, and the probe adds at most 0.06.

Transfer. We also apply the directions learned on 2-hop MuSiQue questions, without retraining, to three other settings (Table 11, Section G). The first is the conversational protocol, in which the model sees its previous answer before the new paragraph. There the directions reach 0.79 (sufficiency), 0.78 (uptake), 0.92 (correction) and 0.82 (stability). This supports the view that the transition is intrinsic to evidence integration, rather than an artefact of the stateless design, in which each context is processed afresh, or of being asked to reconsider an answer. The second is 2WikiMultiHopQA, where

the sufficiency direction reaches 0.96 (0.93 against matched controls), correction 0.85 and uptake 0.79, against 0.99, 0.89 and 0.91 for probes trained on 2WikiMultiHopQA itself. The third is 3–4-hop MuSiQue questions, where sufficiency transfers at 0.82–0.85. Probes trained on the 3–4-hop questions reach 0.93 for sufficiency on every split and score the completion of a leave-one-out context as high as the completion of a chain (Section I). Their uptake readout is less robust: it falls to 0.61 on the document and answer splits, from 0.80 on the question and relation splits.

5.1 What the probes read: state, update and output

Two alternative explanations could account for a high sufficiency AUROC on paired updates. The first is that the update itself is incidental: the information may already be present in the absolute state after (or before) the new evidence, and subtracting the two states would then add nothing. The second is that the probe is simply reading the model's planned output. The anchor is the position from which the first answer token is predicted, and the prompt offers INSUFFICIENT as the alternative to answering. A sufficiency probe could therefore be reading "about to say INSUFFICIENT" rather than any comparison between the evidence and the question. We test both explanations on the same increments, labels and splits.

State versus update. We refit every update probe on four representations of the same increments: the state before the new evidence $h(C_{k-1})$, the state after it $h(C_k)$, their difference Δh, and their concatenation $[h(C_{k-1}); h(C_k)]$ (Table 9, Section D). Most of the sufficiency readout is a property of the state *after* the new evidence. $h(C_k)$ alone reaches 0.930–0.940, against 0.947–0.952 for Δh (matched controls: 0.900–0.908 against 0.910–0.917). The state before the evidence reaches 0.733 in all three models. This is exactly the AUROC of an oracle that knows only the stage at which the increment starts, namely empty, partial or already sufficient: decisive increments make up 20% of the increments that start from a partial context and none of the others. So $h(C_{k-1})$ contributes nothing beyond that stage, and on the matched task, where the decisive increment and its four controls share their starting context, it scores exactly 0.500. The concatenation matches Δh (0.949–0.950). The subtraction therefore adds no information beyond having both endpoints. What it does is remove the question-specific content that both states share, so that a single linear direction works across questions.

For uptake, correction, stability and revision, the state after the evidence is as good as the update or better (uptake 0.769–0.798 against 0.771–0.783; stability 0.813–0.882 against 0.750–0.869). One caveat applies to $h(C_k)$ on the sufficiency task. A detector of "is this context sufficient?" is not enough, because 27% of the negative increments add evidence to a context that is already sufficient. Indeed, the state-sufficiency probe applied to $h(C_k)$ of the same increments scores only 0.70. The update probes solve a different task: "has the context *just* become sufficient?". In short, the model's state represents whether its context has just become sufficient, and the paired update is a clean way to read that. The paired design is what makes the comparison controlled, through matched controls that share a starting context and through the causal tests of Section 6. It is not what makes the information readable.

Evidence versus output. The model's output is itself a sufficiency judgement: it answers when it judges the evidence sufficient and says INSUFFICIENT otherwise. That judgement is often wrong on sufficient evidence. On held-out contexts that contain every gold paragraph, the model says INSUFFICIENT 33% (Qwen), 40% (Llama) and 43% (Mistral) of the time (Table 6). Read as a sufficiency classifier, the output reaches only 0.68–0.76 AUROC. The state-sufficiency probe, read at the anchor of the same forward pass, reaches 0.99. That figure is inflated, because every context with three or more paragraphs is sufficient. We therefore retrain the probe on two-paragraph contexts only, comparing minimal sufficient contexts with every other two-paragraph context, so that paragraph count cannot help. It still reaches 0.976–0.986, against 0.71–0.77 for the model's own answers (Table 7, Section C).

The probe is also accurate where the model's judgement fails. Restricted to the contexts on which the model says INSUFFICIENT, it still separates sufficient from insufficient ones at 0.967–0.982. With its threshold set for 0.95 precision on training data, it recovers 76–81% of the sufficient contexts the model wrongly rejected, with 3–4% false alarms on the insufficient contexts the model correctly rejected. The sufficiency update

probes behave the same way when the output is held fixed: trained and tested only on increments after which the model *still* says INSUFFICIENT, they reach 0.935–0.944, against 0.65–0.69 for the output alone (Table 8). The sufficiency readout is therefore not the planned output. The residual stream registers that the evidence has become complete even in cases where the model's own answer denies it.

The internal signal is weaker in those cases, though. On a scale where the mean probe score of insufficient training contexts is 0 and that of sufficient ones is 1, wrongly rejected contexts score 0.80–0.87, against 1.03 for sufficient contexts the model answers. The internal state is graded, and the verbal decision behaves like a thresholded, noisier read-out of it. This gap between registering complete evidence and acting on it is what uptake measures. It also explains why better sufficiency detection does not by itself buy accuracy in adaptive retrieval (Section O): stopping retrieval at a sufficient context the model will not answer from still yields INSUFFICIENT.

5.2 Is the internal state needed? Text-only baselines

An obvious objection is that a text classifier given the same inputs might do as well as the activation probe. Table 2 trains three such classifiers on the same increments, labels and splits: a TF-IDF logistic regression and two fine-tuned DeBERTa-v3 classifiers (He et al., 2023), one that sees the question and the added paragraph and one that also sees the previous context. For sufficiency, TF-IDF reaches 0.62 and the DeBERTa without context 0.72–0.73. The DeBERTa with context reaches 0.89–0.92 with a reduced training budget (one epoch, 256 tokens, 8k increments) and 0.97 with the full budget (two epochs, 512 tokens, all increments), which exceeds the probe's 0.95 on the same split. Sufficiency is therefore largely recoverable from the text: the activation probe is an accurate readout of a property the input already determines. Uptake is the opposite case. On the same increments no text classifier exceeds 0.63 in predicting whether the model will use the decisive paragraph (Qwen 0.56 with the reduced and 0.63 with the full budget, Llama 0.52, Mistral 0.60, all with context), while the activation probes reach 0.77–0.78. Whether evidence is taken up depends on the model's internal state, not on the evidence alone.

6 Causal results: readout versus mechanism

A direction from which a state can be read is not necessarily a direction the model uses. This section tests whether the probe directions of Section 5 play a causal role in the model's behaviour. All tests use pairs of prompts from 150 held-out questions: an *insufficient* prompt, whose context lacks one gold paragraph, and the corresponding *sufficient* prompt, in which that paragraph has been added. We measure how often the model answers rather than saying INSUFFICIENT, and how often its answer is correct. We use three interventions on the residual stream (Turner et al., 2024; Meng et al., 2022):

- **Steering** adds a multiple of a direction to the residual stream at the anchor and at every generated position. If the direction carried sufficiency, adding it should make the model answer, and subtracting it should make it abstain.
- **Patching** runs the insufficient prompt but replaces its anchor residual with the one from the sufficient prompt, at every layer from the best sufficiency layer onward. Patching only the component of that replacement along a given direction, or only the component orthogonal to it, shows where the causal content lies.
- **Projection-out** runs the sufficient prompt but sets the component of the residual along a direction to its average value on insufficient prompts, from the best layer onward. If the model relies on that direction, it should start abstaining.

We test two kinds of direction: the logistic probe direction v_{lr}, and the difference-of-means direction v_{dm}, the normalised difference between the average residual of sufficient and of insufficient states (Marks and Tegmark, 2023).

Probe directions are inert. Steering along the logistic sufficiency or uptake direction at the best layer changes no aggregate rate at any strength from 0.7 to 22 units of ℓ_2 norm, which is up to 35% of the typical norm of the residual stream (Section L). A random direction of the same norm is equally inert, so the null result is not a matter of dose. Patching only the component of the sufficient-minus-insufficient update that lies along the probe

Model	Budget	Task	TF-IDF	DeBERTa (q+added)	DeBERTa (+context)	Probe
Qwen2.5-7B	reduced budget	sufficiency	0.623	0.723	0.900	0.948
Qwen2.5-7B	reduced budget	sufficiency matched	0.623	0.758	0.823	0.911
Qwen2.5-7B	reduced budget	uptake	0.574	0.569	0.557	0.775
Qwen2.5-7B	full budget	sufficiency	0.611	0.723	0.972	0.948
Qwen2.5-7B	full budget	uptake	0.574	0.625	0.630	0.775
Llama-3.1-8B	reduced budget	sufficiency	0.625	0.715	0.888	0.952
Llama-3.1-8B	reduced budget	uptake	0.463	0.466	0.517	0.783
Mistral-7B	reduced budget	sufficiency	0.625	0.730	0.920	0.949
Mistral-7B	reduced budget	uptake	0.536	0.573	0.604	0.771

Table 2: Text-only classifiers trained on the same increments and splits as the activation probe (document split). "q+added": the classifier sees the question and the added paragraph; "+context": it also sees the previous context.

	Patching the insufficient prompt (correct/answered)								Projection-out on sufficient (answered)		
Model	none	full	$\parallel v$	$\perp v$	rand-256	PC1	PC256	unemb.	v_{dm}	random	v_{lr}
Qwen2.5-7B	0.04/0.13	0.26/0.71	0.04/0.13	0.25/0.71	0.04/0.13	0.09/0.29	0.18/0.59	0.09/0.19	0.01	0.72	0.72
Llama-3.1-8B	0.02/0.11	0.24/0.74	0.02/0.12	0.24/0.73	0.02/0.12	0.02/0.19	0.09/0.41	0.19/0.62	0.00	0.73	0.75
Mistral-7B	0.07/0.17	0.09/0.60	0.07/0.17	0.09/0.60	0.07/0.17	0.08/0.61	0.08/0.56	0.07/0.19	0.01	0.63	0.63
Qwen 3–4 hop	0.05/0.20	0.25/0.62	0.06/0.19	0.25/0.62	0.06/0.20	0.07/0.23	0.20/0.51	0.07/0.21	0.03	0.63	0.63

Table 3: Causal tests on 150 held-out questions, intervening at every layer from the best sufficiency layer onward. Left: the insufficient prompt with its anchor residual patched from the sufficient prompt (correct / answered rate). "none": no patch; "full": the whole residual; $\parallel v$ and $\perp v$: only the component along, or orthogonal to, the probe direction; "rand-256": a random 256-dimensional subspace; PC1, PC256: the top 1 or 256 principal components of training updates; "unemb.": the gold answer's unembedding direction. Right: the sufficient prompt with the component along a direction set to its insufficient-state value (answered rate). v_{dm}: difference-of-means direction; v_{lr}: logistic-probe direction.

direction leaves the insufficient prompt's behaviour unchanged in all three models (Table 3, "$\parallel v$"). Projecting the logistic direction out of sufficient prompts changes nothing either (v_{lr} column). The same holds for the uptake direction.

The anchor residual is causally sufficient, but its effect lies outside the probe direction. Replacing the insufficient prompt's anchor residual with the sufficient prompt's takes Qwen from $0.04/0.13$ (correct/answered) to $0.26/0.71$ and Llama from $0.02/0.11$ to $0.24/0.74$. This is roughly half the correctness the model reaches when the missing paragraph is actually present. Removing the probe component from that patch ("$\perp v$") leaves the effect intact ($0.25/0.71$, $0.24/0.73$). The single anchor position therefore carries enough to make the model answer, and often answer correctly, but none of that content lies along the direction the probe reads. Mistral shows the same dissociation with a weaker effect of the full patch ($0.09/0.60$), as does the 3–4-hop run ($0.25/0.62$).

The causal content is high-dimensional and model-specific. To measure how many dimensions the causal content occupies, Fig. 3 patches only the part of the update that lies in the top-k principal components of the decisive updates seen in training. The answering rate rises gradually with k (Qwen: 0.29 at k=1, 0.49 at k=16, 0.59 at k=256, 0.71 for the full update), while a random 256-dimensional subspace does nothing. Correctness needs even more of the update. The first component carries much of the switch between answering and abstaining, which is consistent with the projection-out result below. What carries correctness differs by model. One natural candidate is the gold answer's unembedding direction, the direction in the residual stream that most directly promotes the answer's first token. On Llama this direction alone restores $0.19/0.62$, most of the full effect. On Qwen and Mistral it restores little ($0.09/0.19$, $0.07/0.19$), although removing it from the full patch halves Qwen's correctness ($0.26 \rightarrow 0.13$). Mistral's and the 3–4-hop run's effects are captured by the top principal components instead (0.56 and 0.51 answered at k=256).

The mean-difference direction is a causal abstention gate. The one direction with a causal effect is v_{dm}, which the logistic probe does not align with. Setting its component to the insufficient-state value makes all three models abstain on essentially every sufficient prompt (answered 0.01, 0.00, 0.01, against 0.72, 0.73, 0.63 for a random direc-

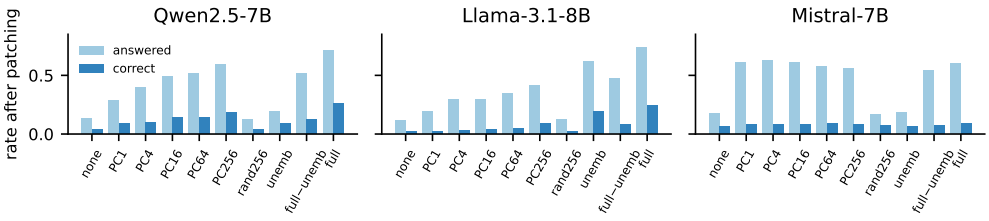


Figure 3: How many dimensions the causal content needs. The insufficient prompt is patched with the sufficient-minus-insufficient update projected on the top-k principal components of training updates, on a random 256-dimensional subspace, on the gold answer's unembedding direction ("unemb"), or on everything but that direction.

tion treated identically), and equally on prompts with falsified evidence. Steering along v_{dm} at low strength over four layers gives a graded, monotone response in both directions, while the random control stays flat (Fig. 4; full numbers in Table 12, Section M). Here the strength α is measured in units of the gap between the sufficient and insufficient means. At $\alpha = -1$ Qwen answers 34% of sufficient prompts (from 71%), with unchanged precision among the answers it still gives.

The positive direction, however, is indiscriminate. At $\alpha = +1$ the models answer more often on distractor-only prompts (Qwen $0.03 \rightarrow 0.12$, Llama $0.03 \rightarrow 0.29$, Mistral $0.03 \rightarrow 0.28$), on insufficient prompts ($0.13 \rightarrow 0.26$, $0.11 \rightarrow 0.51$, $0.17 \rightarrow 0.47$) and on answer-string controls, and Llama follows falsified evidence twice as often ($0.19 \rightarrow 0.39$). Meanwhile, correct answers on sufficient prompts gain only 3 points on Qwen and 18 on Llama. Before running these experiments we fixed a criterion: a direction that makes the model accept any passage is a context-compliance or abstention direction, not one that makes the model integrate evidence better. By that criterion, v_{dm} is an abstention gate. Applied at all positions and all layers with $|\alpha| \geq 2$, the same recipe destroys the output for the probe and random directions alike (Section M).

Use for adaptive retrieval. Because the readouts are available before generation, they can decide when a retrieval system should stop fetching evidence and answer. Used as such a stopping rule, the sufficiency readouts match the behavioural rule "stop as soon as the model does not say INSUFFICIENT" on MuSiQue and beat it on HotpotQA (0.73 against 0.65 accuracy), without generating at every round. Uptake, correction and stability are poor stopping rules. Once retrieval has stopped, however, the uptake (or correction) direction is the best learned signal for deciding whether to trust the answer given there (Section O).

7 External validity: other benchmarks, natural retrieval, domain shift

The results so far come from MuSiQue, a dataset whose questions were composed from single-hop questions over curated Wikipedia paragraphs. We now ask whether they depend on that construction. We repeat the whole pipeline, unchanged and with Qwen2.5-7B, on three further sources (Table 4):

- **HotpotQA** (Yang et al., 2018) was crowdsourced to test multi-hop reasoning over pairs of Wikipedia paragraphs, with annotated supporting sentences so that models can be asked to explain their answers. It contains both *bridge* questions, where the first paragraph leads to the entity the second one describes, and *comparison* questions, whose two facts are independent lookups. We use 2,000 questions from its distractor setting, taking the two supporting paragraphs as gold hops and the provided distractor paragraphs as distractors. Unlike in the main runs, we keep questions the model can answer closed-book, so that results can be split by closed-book status.

- **RAGBench** (Friel et al., 2024) was built to evaluate RAG systems on realistic inputs. It collects twelve existing question-answering datasets from biomedical, finance, technical, customer-support and general domains, together with the documents a real retrieval pipeline returned for each question, and annotates which sentences of those documents support the reference answer. We use 3,000 of its mostly single-hop questions. The documents with annotated relevant sentences are the gold evidence and the other retrieved documents are the distractors, so here the distractors are

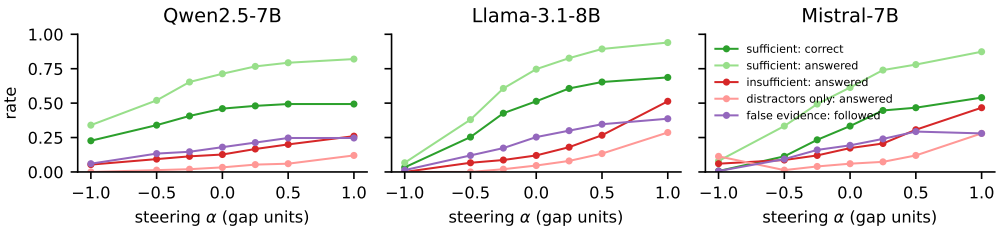


Figure 4: Dose-response of difference-of-means steering (four layers from the best sufficiency layer, from the anchor onward; α in units of the sufficient-minus-insufficient gap). Negative α induces abstention everywhere; positive α induces answering everywhere, including on distractors and false evidence.

naturally retrieved rather than curated. RAGBench provides long-form reference answers, so we judge correctness with an LLM judge that compares the model's answer with the reference.

- **A synthetic benchmark**, which we generate, provides full control over the evidence. Its 2,400 questions concern fictional entities, so the model cannot know any answer closed-book (its closed-book accuracy is zero). The questions require one to three hops over seven relations, each expressed with five template families. Every gold fact also comes in a paraphrased, a partial (value withheld) and a contradictory (different value) version, and the distractors include near misses such as the same relation for another entity.

Readouts. Sufficiency is decodable on every source (0.98 HotpotQA, 0.90 RAGBench, 0.97 synthetic) and against matched controls (0.96, 0.94, 0.95). On the synthetic data, these controls include paraphrased duplicates, partial evidence and contradictions. Projected onto the sufficiency direction, the synthetic increments are ordered as on MuSiQue: decisive (whether original, paraphrased or completing a leave-one-out context) $>$ contradictory $>$ partial $\approx$ duplicate $>$ distractor $>$ answer-string control. Uptake is also decodable on all three sources (0.73, 0.69, 0.93). Breaking results down by question group (Table 15, Section P) shows that uptake is not an artefact of multi-hop questions. It reaches 0.70 on RAGBench single-hop questions (n=444), 0.79 on HotpotQA comparison questions against 0.68 for bridge questions, and 0.99 on single-hop synthetic questions. It is above chance in every RAGBench domain (0.56–0.78). On HotpotQA it is above chance both for questions the model answers correctly closed-book (0.80) and for those it does not (0.72). Sufficiency is decoded equally well in both groups (0.984 against 0.978), so the readout is not a signal that the model "needs" evidence it lacks.

Transfer across construction and domain. The MuSiQue directions transfer to HotpotQA at 0.95 for sufficiency, but only partially to RAGBench (0.71; matched 0.78; uptake 0.55) and to the synthetic data (0.74; uptake 0.72; correction 0.80). Within RAGBench, holding out whole sub-datasets at test time drops sufficiency from 0.90 to 0.72 (matched 0.94 $\rightarrow$ 0.87, uptake 0.69 $\rightarrow$ 0.60). Holding out relation chains on the synthetic data costs two to five points. The state is therefore present under natural retrieval and in every domain, but its direction is partly dataset-specific: a detector trained on Wikipedia should not be deployed without adaptation on biomedical or technical RAG.

Causal dissociation. The dissociation of Section 6 replicates on all three sources. The full anchor patch makes the insufficient prompt answer (0.41 $\rightarrow$ 0.92, 0.15 $\rightarrow$ 0.72, 0.03 $\rightarrow$ 0.89), the component along the probe direction does nothing, and the orthogonal component carries the whole effect. Projecting out v_{dm} silences answering (0.17, 0.10, 0.00), while a random direction does not. On synthetic data the patched model answers 89% of the time but is never correct. The anchor carries the decision to answer, while a fictional answer can only be read from the evidence tokens, which the insufficient prompt lacks.

Evidence State Transition Benchmark: retrieval source and linguistic form. Finally, we vary how the evidence itself is obtained and worded, while keeping the MuSiQue questions. In a real RAG system, evidence comes from a retriever rather than from a benchmark's curated paragraphs. We therefore rebuild the trajectories of 1,200 MuSiQue ques-

	within-dataset probes (AUROC)				MuSiQue dir.		patched insufficient (answered)				proj.-out (answered)	
Dataset	suff.	matched	uptake	corr.	suff.	uptake	none	full	$\parallel v$	$\perp v$	v_{dm}	random
HotpotQA	0.979	0.961	0.729	0.846	0.952	0.649	0.41	0.92	0.41	0.92	0.17	0.92
RAGBench	0.898	0.944	0.685	0.742	0.706	0.550	0.15	0.72	0.15	0.72	0.10	0.73
RAGBench (held-out sub-datasets)	0.717	0.872	0.596	0.719	0.706	0.550	–	–	–	–	–	–
Synthetic	0.965	0.948	0.930	0.925	0.739	0.722	0.03	0.89	0.03	0.89	0.00	0.99
Synthetic (held-out relation chains)	0.945	0.925	0.883	0.897	0.739	0.722	–	–	–	–	–	–

Table 4: External validity (Qwen2.5-7B). Within-dataset probes trained and tested on the row's dataset; MuSiQue 2-hop directions applied without retraining; and the patching and projection-out tests of Section 6 (answered rate, 150 held-out questions). "Held-out sub-datasets" and "held-out relation chains" test on groups never seen in training.

tions with passages retrieved from the 21M-passage Wikipedia corpus distributed with FlashRAG (Jin et al., 2025; Karpukhin et al., 2020), using either BM25 (Robertson and Zaragoza, 2009), a lexical retriever, or E5 (Wang et al., 2022), a dense one. Each question issues the question itself plus one query per hop. In one setting the gold paragraphs are kept and all distractors are retrieved. In the other, *all* evidence is retrieved, which is possible for the 402 (E5) and 219 (BM25) questions for which retrieval found every hop. A further run varies the wording of the decisive paragraph: it adds LLM-paraphrased and value-withheld (partial) versions of each gold paragraph, and paragraphs on the same entity written independently for HotpotQA or 2WikiMultiHopQA. Nothing changes qualitatively (Table 19; construction, measurements and full results in Section Q). Sufficiency stays at 0.91–0.95. The MuSiQue directions transfer at 0.91–0.94 (sufficiency) and 0.76–0.82 (uptake) on questions the source probes never saw, and the causal dissociation and abstention gate reproduce in every setting. Paraphrased and independently written decisive paragraphs project like the originals. Partial and contradictory versions are the hardest controls, separated from decisive increments at only 0.81 and 0.76 (Table 20).

8 Discussion

Two readable signals and one causal gate. Our results separate three things that the phrase "an internal state transition" tends to conflate. *Sufficiency* is a property of the input: whether the context now contains every fact the question needs. The model represents it cleanly, but a text classifier that sees the same context can compute it too, and its causal counterpart in the model is an abstention gate. *Uptake* is a property of the model: whether the same decisive paragraph will actually be used in the answer. It is predictable from activations but not from text, at 0.78 AUROC across three model families and on every external source. Neither readout direction is a lever that controls the behaviour it predicts. What turns an insufficient state into an answer is high-dimensional, orthogonal to both readout directions, and on some models concentrated on the answer's unembedding direction. The picture is therefore a low-dimensional gate that decides whether to answer, plus high-dimensional content that determines the answer, rather than a single direction along which the model takes up new evidence.

Why probes do not find the lever, and what to use instead. A logistic probe looks for the direction along which the two classes are best separated relative to their spread. It therefore down-weights directions along which activations vary a lot within each class. The difference-of-means direction, in contrast, follows the raw shift between the class averages, and much of that shift lies along a high-variance axis related to answering. The projection-out and subspace experiments show that the causal content lies exactly there. A probe direction can thus score a high AUROC and transfer well across datasets while being causally irrelevant. Any proposed steering direction should therefore be validated with controls of the kind used here: projecting the direction out, measuring how many dimensions the effect needs, and comparing with random directions of matched norm. For retrieval systems, the most useful signal is uptake, as a detector of evidence the model has ignored. Sufficiency is tracked about as well by the model's own abstentions, and the abstention gate trades abstention against compliance uniformly rather than making the model more sensitive to the evidence.

9 Conclusion

We compared the model's internal state before and after each new piece of evidence, using evidence trajectories with matched controls. Evidence sufficiency is readable before generation, transfers

across datasets, and can also be recovered from the text. Evidence uptake is readable and transfers, but cannot be recovered from the text. Neither readout direction causes the behaviour it predicts; the one causal direction we found is an abstention gate. We release the trajectory construction, probing and intervention code, so that checking whether a readout is also a mechanism can become standard practice in studies of evidence-aware generation.

Limitations

The causal experiments on the external sources use Qwen only. RAGBench correctness relies on an LLM judge. Directions learned on Wikipedia transfer only partially to other domains. Uptake weakens on 3–4-hop questions under the document-disjoint split (0.61). Finally, we tested only additive, patching and projection interventions at the anchor position.

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A Evidence conditions and increments

This appendix lists every context (*condition*) shown to the model for one question, and every pair of conditions (*increment*) on which probes are trained. Consider a question with n gold paragraphs $g_0, \ldots, g_{n-1}$, one per reasoning hop, in the order given by the dataset's decomposition of the question, together with its pool of distractor paragraphs. The conditions are:

- **Empty and distractor-only contexts.** C_0 contains no evidence. C_D contains n distractors, each matched in word count to one gold paragraph.
- **Chains.** Gold paragraphs are added one at a time, in the canonical hop order, in the reverse order and, for $n \geq 3$, in random orders. Each chain gives the prefixes $C_{o_1}, C_{o_1 o_2}, \ldots$, and its last prefix is the minimal sufficient context. Each new paragraph is inserted at a uniformly random position among the paragraphs already present, so that the position of the decisive paragraph carries no information. Identical contexts reached through different orders are stored once.
- **Leave-one-out contexts** ($n \geq 3$): all gold paragraphs but one, shuffled, followed by their completion with the missing paragraph.
- **Matched controls.** From every *penultimate* context S, i.e. a chain or leave-one-out context that lacks exactly one gold paragraph, we build four controls alongside the decisive completion: $S{+}D$ adds a distractor matched in length to the missing paragraph; $S{+}R$ adds a verbatim duplicate of a gold paragraph already in S; $S{+}F$ adds the missing paragraph with its sub-answer (and title, if applicable) replaced by another question's sub-answer for the same relation, so the false paragraph remains plausible in type (questions whose sub-answer does not appear in the paragraph have no such control); $S{+}A$ adds a distractor with the sentence "(Not to be confused with {answer}.)" appended, so the answer string is present without the supporting fact.
- **Post-sufficiency contexts.** The canonical minimal sufficient context plus a distractor, plus a duplicate gold paragraph, or plus a falsified copy of the final hop that conflicts with it; and the full context of all 20 paragraphs, shuffled.

For 2-hop questions this gives 17 conditions and 17 increments per question. Table 5 names the kinds of increment, as they appear in the tables and figures, and gives their sufficiency label. The *sufficiency* task uses all kinds. The *matched* variant uses only decisive increments and the four controls built from the same penultimate context. The conversational protocol (Section H) re-runs the 10 chain-related increment kinds with the model's own stateless answer shown in the prompt (30,000 increments for Qwen).

B Probing, nuisance and intervention details

Activation capture. Activations are read at the anchor position by forward hooks on the output of each decoder block. Prompts are processed in batches, left-padded so that the anchor has the same index in every row. In the same forward pass we append the gold answer tokens after the anchor. Because attention is causal, this leaves the anchor's activations unaffected, but it gives the log-probability the model assigns to the gold answer, which we use as a nuisance covariate. The cached keys and values are then cut back to the prompt, and the model decodes greedily from there. Layer index l denotes the residual stream after block l ($l = 0$: embeddings).

Probes. Probes are `scikit-learn` logistic regressions with L2 regularisation ($C = 0.5$) and balanced class weights, on features standardised on the training split (Alain and Bengio, 2016); the layer is selected by grouped 3-fold cross-validation

Kind	What the increment adds	y^{suff}
`decisive`	the last missing gold paragraph of a chain	1
`loo_decisive`	the missing gold paragraph of a leave-one-out context	1
`gold_nondecisive`	a gold paragraph that still leaves another one missing	0
`distractor_from_none`	distractors added to the empty context	0
`distractor`	a length-matched distractor added to a penultimate context ($S{+}D$)	0
`redundant`	a duplicate of a gold paragraph already present ($S{+}R$)	0
`false`	the missing paragraph with a substituted sub-answer ($S{+}F$)	0
`answer_ctrl`	a distractor that mentions the answer string ($S{+}A$)	0
`post_distractor`	a distractor added to the minimal sufficient context	0
`post_redundant`	a duplicate gold paragraph added to the minimal sufficient context	0
`post_false`	a falsified copy of the final hop, conflicting with it, added to the minimal sufficient context	0
`full_context`	all remaining paragraphs added to the minimal sufficient context	0

Table 5: Kinds of increment and their sufficiency label y^{suff} (1 when the increment makes the context sufficient for the first time).

on training questions (never on test), and the reported number is the test AUROC at that layer. The probe direction is the weight vector w divided by the per-feature standard deviations σ, so that it lives in the original activation space, and normalised to unit norm. The difference-of-means direction is the normalised difference of class means of Δh (or of h for state tasks). Bootstrap intervals resample test questions (1,000 draws) using the stored probe weights.

State probes. State probes use the stored anchor activation of each condition, one example per condition, with the same estimator, standardisation, layer selection and question-level splits as the update probes. For a two-hop question the state-sufficiency task has five positive conditions: the minimal sufficient context in each hop order (de-duplicated when identical), sufficient plus a distractor, sufficient plus a duplicate, and the full context. It has ten negative ones: no evidence, distractors only, each single hop, and each penultimate state plus a distractor, a duplicate or an answer-string control. Conditions containing a falsified paragraph are excluded. All positives except the minimal sufficient context contain three or more paragraphs, and no negative does, so the task is partly separable by paragraph count; Section C reports a count-matched version restricted to two-paragraph contexts. State-correctness and abstention-flag probes use every condition, labelled by the model's own answer.

Nuisance covariates. Δtokens, total tokens, Δ gold log-probability, Δ next-token entropy, Δ top probability, answer string in added text / in the full context / in the previous context, TF-IDF cosine relevance of the added passage to the question, number and word count of added paragraphs, relative insertion position, and a 16-dimensional truncated-SVD embedding of the added text's TF-IDF vector (fit on training increments). The nuisance-only classifier is a logistic regression on these covariates. The residualised probe first regresses Δh on the covariates with ridge regression ($\lambda = 1$) and then probes what the covariates do not explain. The "gain" is the AUROC added when out-of-fold probe scores (5-fold, grouped by question) are given to the nuisance classifier as an extra feature.

Text baselines. TF-IDF (1–2-grams, sublinear tf, $\leq$ 200k features) with logistic regression; DeBERTa-v3-base fine-tuned with AdamW (lr $3{\times}10^{-5}$, linear decay, class-weighted cross-entropy). The reduced budget uses one epoch, 256 tokens and 8,000 training increments (all positives kept); the full budget two epochs, 512 tokens and all training increments.

Interventions. Steering adds the vector $\alpha\, u\, v$ at the anchor and every generated position ("anchor onward") or at all positions; the unit u is $|\mathbb{E}[v^\top h \mid \text{sufficient}] - \mathbb{E}[v^\top h \mid \text{insufficient}]|$ on training conditions, computed per layer, so that $\alpha = 1$ moves the activation by one gap between the two class means. Projection-out sets $v^\top h$ to the insufficient-state mean at the anchor and generated positions for every layer from the best layer onward. Patching replaces the anchor residual during the prefill at every layer from the best layer to the last; decoding then proceeds from the patched cache. Principal components are computed from 1,500 training decisive updates per layer; the "unembedding" direction is the normalised row of the

output embedding matrix for the gold answer's first token.

C Behaviour, verbalised sufficiency and the internal state

This appendix does three things. It describes how the models behave on each condition family, which supplies the behavioural labels behind uptake, correction, stability and revision. It shows that the model's own output, read as a judgement about whether the evidence is sufficient, is frequently wrong. And it shows that the sufficiency state read from the residual stream at the same position, in the same forward pass, is a much more accurate signal than that verbalised judgement, including in the cases where the verbalised judgement is wrong.

C.1 Behaviour by condition family

Table 6 reports, for every condition family of Section A, the fraction of greedy answers that are correct and the fraction that are INSUFFICIENT. It covers the evidence-only regime over the 2,000 two-hop questions of each model. The remaining fraction is a wrong answer. All questions are ones the model answers incorrectly closed-book (Section 3), so a correct answer can be attributed to the evidence.

Read row by row:

- **No evidence and distractors only.** The instruction is followed. The models almost never answer (94–100% INSUFFICIENT) and are never correct. With distractors alone, the 3–6% of prompts that are answered are hallucinations.

- **One hop.** With only one of the two gold paragraphs, the evidence is insufficient by construction, yet the models answer 20–31% of the time and are correct 10–13% of the time. Some second hops can be guessed from the first, which is one reason correctness and sufficiency are separate labels.

- **Minimal sufficient.** With both gold paragraphs present and nothing else, the models answer correctly only 36–47% of the time, and they say INSUFFICIENT 30–36% of the time. The evidence is sufficient by construction: MuSiQue composes each question from single-hop questions whose supporting paragraphs are exactly these gold paragraphs. An INSUFFICIENT output here is therefore a *verbalised false negative*: the model declares the evidence insufficient when it is not. The remaining 22–28% are wrong answers.

- **One hop plus a falsified final hop.** Replacing the decisive paragraph with a version whose sub-answer is swapped roughly halves abstention (49–55% INSUFFICIENT, against 69–80% with one hop alone). The models treat a type-plausible but false final hop much as they treat the true one, which is what the sufficiency readout also does (Section 5: the false increment projects second only to the decisive one).

- **Sufficient plus a distractor, and the full context.** Adding irrelevant paragraphs to a sufficient context makes the models abstain *more*: 35–47% with one extra distractor and 38–61% with all 20 paragraphs. Correctness falls accordingly (0.17–0.39). Verbalised sufficiency judgements degrade as the context grows, even though the evidence is unchanged.

C.2 The verbalised judgement is a poor sufficiency detector; the internal state is a good one

Because the prompt instructs the model to output INSUFFICIENT exactly when the evidence does not determine the answer, its output is a binary sufficiency judgement: *answers* means "judged sufficient", *INSUFFICIENT* means "judged insufficient". We compare this verbalised judgement with the state-sufficiency probe (Section 4), which reads the residual stream at the answer-start anchor of the same forward pass, before any token is generated.

Protocol. The comparison uses the held-out conditions of the document split, with conditions containing falsified evidence excluded, as in the probe task. These are about 2,700 sufficient and 6,000 insufficient contexts per model. For the verbalised judgement we report its true-positive rate (sufficient contexts answered), its false-positive rate (insufficient contexts answered) and its AUROC as a binary score. For the probe we report AUROC and two operating points, both with the threshold set on *training* contexts. The first matches the false-positive rate of the verbalised judgement, so the probe raises false alarms on insufficient contexts as often as the model's own answers do. The second requires 0.95 precision. At each operating point

Condition (correct / abstain)	Qwen2.5-7B	Llama-3.1-8B	Mistral-7B
no evidence	0.00 / 1.00	0.00 / 1.00	0.00 / 0.99
distractors only	0.00 / 0.97	0.00 / 0.97	0.00 / 0.94
one hop	0.10 / 0.80	0.12 / 0.76	0.13 / 0.69
minimal sufficient	0.46 / 0.30	0.47 / 0.31	0.36 / 0.36
one hop + false final hop	0.13 / 0.49	0.13 / 0.55	0.12 / 0.51
sufficient + distractor	0.39 / 0.35	0.38 / 0.41	0.27 / 0.47
full context (20 par.)	0.22 / 0.38	0.17 / 0.61	0.17 / 0.48

Table 6: Behaviour by condition family (evidence-only regime, 2,000 questions per model): fraction correct / fraction answering INSUFFICIENT. The remaining fraction is a wrong answer.

we report the probe's recall on the sufficient contexts the model wrongly called INSUFFICIENT (*wrong abstentions*), and its false-alarm rate on the insufficient contexts the model correctly called INSUFFICIENT.

A confound, and how we remove it. Every condition with three or more paragraphs is sufficient (sufficient plus a distractor, sufficient plus a duplicate, full context). Every insufficient condition has at most two. Paragraph count, and with it prompt length, therefore separates part of the state task on its own. This inflates the state probe's headline AUROC (0.99) and makes its recall on wrong abstentions in the three larger families trivially perfect. We therefore also report a *paragraph-count-matched* version. It keeps only two-paragraph contexts: the minimal sufficient context against distractors only and the penultimate state plus a distractor, a duplicate or an answer-string control. The probe is retrained on these contexts alone, so neither count nor length can separate the classes. The count-matched numbers are the ones we interpret.

Results (Table 7).

- **The verbalised judgement is wrong on a third or more of sufficient contexts.** On held-out data the models call 33% (Qwen), 40% (Llama) and 43% (Mistral) of sufficient contexts INSUFFICIENT, while still answering 15–21% of insufficient ones. As a sufficiency classifier the output reaches 0.76, 0.72 and 0.68 AUROC (0.77, 0.77, 0.71 on two-paragraph contexts).

- **The internal state is far more accurate, with paragraph count held fixed.** On two-paragraph contexts the retrained probe separates minimal sufficient from insufficient contexts at 0.976, 0.981 and 0.986 AUROC. This is 0.20–0.27 above the verbalised judgement made in the same forward pass.

- **It is accurate precisely where the verbalised judgement fails.** Restricted to the two-paragraph contexts the model called INSUFFICIENT, so that the output is held fixed, the probe still separates the sufficient ones from the insufficient ones at 0.967–0.982. At 0.95 training precision it recovers 76%, 77% and 81% of the wrongly rejected minimal sufficient contexts, with 3–4% false alarms on correctly rejected insufficient ones. At the verbalised false-alarm rate it recovers 96–99% of them. In practical terms, a detector reading the anchor before generation would accept most of the complete evidence the model itself refuses, at a fraction of the false alarms the model's own answers produce.

- **But the internal state is weaker in exactly those cases.** In units where the insufficient training mean is 0 and the sufficient one is 1, wrongly rejected minimal sufficient contexts score 0.82, 0.80 and 0.87, against 1.03 for sufficient contexts the model answers. Insufficient contexts score about 0 when rejected and 0.08–0.16 when (wrongly) answered. The internal sufficiency state is graded. The verbal decision behaves like a thresholded and noisier read-out of it, which tends to fail when the state is weaker than usual, but the state in those cases is still much closer to "sufficient" than to "insufficient". (Without the count matching, the wrongly rejected contexts appear to score as high as answered ones, 1.03–1.15. That impression comes from the larger, trivially separable families and should not be relied on.)

- **Adding paragraphs hurts the verbal judgement.** The abstention rate on sufficient contexts rises from 29–37% (minimal) to 40–60% (full context), while every wrongly rejected full-context prompt is still on the sufficient

	Qwen2.5-7B	Llama-3.1-8B	Mistral-7B
Held-out contexts			
sufficient / insufficient	2754 / 6090	2701 / 6000	2754 / 6100
sufficient but verbalised INSUFFICIENT	901 (33%)	1079 (40%)	1183 (43%)
Verbalised judgement (answers = sufficient)			
AUROC	0.762	0.722	0.681
TPR / FPR	0.673 / 0.150	0.601 / 0.156	0.570 / 0.208
State-sufficiency probe at the same anchor			
AUROC	0.992	0.994	0.996
TPR / FPR, threshold at verbalised FPR	0.995 / 0.188	0.996 / 0.178	1.000 / 0.194
recall on wrongly abstained sufficient contexts	0.990	0.991	0.999
false alarms on correctly abstained insufficient contexts	0.154	0.151	0.167
TPR / FPR, threshold at 0.95 precision	0.962 / 0.050	0.970 / 0.051	0.977 / 0.038
recall on wrongly abstained sufficient contexts	0.936	0.956	0.961
false alarms on correctly abstained insufficient contexts	0.039	0.040	0.029
Wrong abstentions by condition: abstain rate; probe recall at 0.95 precision			
minimal sufficient	0.29; 0.79	0.30; 0.84	0.37; 0.87
sufficient + distractor	0.35; 1.00	0.41; 1.00	0.49; 1.00
sufficient + duplicate	0.29; 0.99	0.34; 0.99	0.39; 1.00
full context (20 par.)	0.40; 1.00	0.60; 1.00	0.50; 1.00
Mean probe score (0 = insufficient mean, 1 = sufficient mean on training contexts)			
sufficient, answered	0.98	0.88	0.95
sufficient, verbalised INSUFFICIENT	1.03	1.15	1.04
insufficient, answered	0.10	0.08	0.04
insufficient, verbalised INSUFFICIENT	-0.01	-0.00	-0.03
Two-paragraph contexts only (paragraph count matched; probe retrained on them)			
minimal sufficient / insufficient two-paragraph	927 / 4263	901 / 4200	924 / 4270
minimal sufficient but verbalised INSUFFICIENT	268	273	341
verbalised judgement: AUROC; TPR / FPR	0.774; 0.71 / 0.16	0.773; 0.70 / 0.15	0.711; 0.63 / 0.21
probe AUROC (all two-paragraph contexts)	0.976	0.981	0.986
probe AUROC among contexts verbalised INSUFFICIENT	0.967	0.972	0.982
threshold at verbalised FPR: TPR / FPR	0.984 / 0.196	0.986 / 0.187	0.997 / 0.213
recall on wrong abstentions; false alarms on correct abstentions	0.963; 0.171	0.963; 0.161	0.991; 0.188
threshold at 0.95 precision: TPR / FPR	0.859 / 0.041	0.890 / 0.047	0.905 / 0.039
recall on wrong abstentions; false alarms on correct abstentions	0.757; 0.032	0.773; 0.037	0.812; 0.028
mean probe score: sufficient answered / sufficient abstained	1.03 / 0.82	1.03 / 0.80	1.03 / 0.87
mean probe score: insufficient answered / insufficient abstained	0.16 / -0.01	0.14 / -0.01	0.08 / -0.04

Table 7: Verbalised versus internal sufficiency judgements on held-out contexts (document split; falsified-evidence contexts excluded). Verbalised: the model answers (judged sufficient) or says INSUFFICIENT. Probe: state-sufficiency probe at the answer-start anchor of the same forward pass; thresholds are set on training contexts. "Wrong abstentions": sufficient contexts the model called INSUFFICIENT; "correct abstentions": insufficient contexts it called INSUFFICIENT. Mean probe scores are expressed so that 0 is the mean of insufficient and 1 the mean of sufficient training contexts. Top block: all contexts, inflated by paragraph count. Bottom block: two-paragraph contexts only, with the probe retrained on them.

side of the probe's threshold. Paragraph count makes that last comparison easy, so we do not interpret it further.

C.3 Holding the output fixed for the update probes

The same question can be asked of the update probes of Table 1: is "the evidence just became sufficient" anything more than "the model just switched from INSUFFICIENT to an answer"? Table 8 reports, for the sufficiency and matched-sufficiency tasks with Δh and with $h(C_k)$ (Section D), and for the state-sufficiency task, four numbers:

- the AUROC of the output alone: the indicator that the model answers at C_k (for updates, the indicator that it switched from INSUFFICIENT to an answer scores within 0.04 of this);
- the standard probe, trained on all increments, scored separately on the test increments after which the model *still says INSUFFICIENT* and on those after which it answers ("stratified");
- probes trained *and* tested only within each of these two groups ("within"). Within the abstaining group, the positives are decisive increments that completed the evidence while the model went on refusing, and the negatives are controls after which it also refused.

The output alone gives 0.65–0.76. The probes lose almost nothing when the output is held fixed. Among increments after which the model still says INSUFFICIENT, the Δh sufficiency probe scores 0.940–0.949 (stratified) and 0.935–0.944 (within), against 0.945–0.950 overall; for $h(C_k)$ the corresponding figures are 0.930–0.944 and 0.914–0.933. Among increments after which the model answers,

			output	all	model abstains at C_k		model answers at C_k	
Model	Task	Repr.	only	test	stratified	within	stratified	within
Qwen2.5-7B	Sufficiency	Δh	0.685	0.945	0.948	0.944	0.929	0.927
	Sufficiency	$h(C_k)$	0.685	0.930	0.930	0.918	0.904	0.890
	Matched	Δh	0.721	0.911	0.891	0.884	0.879	0.870
	Matched	$h(C_k)$	0.721	0.909	0.904	0.883	0.854	0.843
	State suff.	$h(C)$	0.762	0.992	0.990	0.982	0.986	0.983
Llama-3.1-8B	Sufficiency	Δh	0.692	0.950	0.940	0.935	0.942	0.929
	Sufficiency	$h(C_k)$	0.692	0.927	0.931	0.914	0.893	0.892
	Matched	Δh	0.726	0.912	0.896	0.883	0.875	0.855
	Matched	$h(C_k)$	0.726	0.908	0.908	0.886	0.840	0.838
	State suff.	$h(C)$	0.722	0.992	0.992	0.988	0.986	0.982
Mistral-7B	Sufficiency	Δh	0.646	0.948	0.949	0.937	0.939	0.931
	Sufficiency	$h(C_k)$	0.646	0.936	0.944	0.933	0.912	0.912
	Matched	Δh	0.671	0.910	0.918	0.887	0.874	0.876
	Matched	$h(C_k)$	0.671	0.902	0.909	0.872	0.860	0.856
	State suff.	$h(C)$	0.681	0.996	0.996	0.994	0.994	0.989

Table 8: Sufficiency probes with the model's output held fixed (document split). Output only: AUROC of the indicator "the model answers at C_k". All: standard probe, all test increments. Stratified: the same probe on the test increments after which the model abstains (resp. answers). Within: probe trained and tested only on that group, layer re-selected by grouped inner cross-validation. The state-sufficiency rows are not count-matched (see Table 7 for the count-matched version).

the probes score slightly lower (0.929–0.942 for Δh), because there the controls that the model answers (false evidence, answer-string controls) are the hardest ones. On the matched task, where every control starts from the same state as the decisive increment, the probes reach 0.883–0.887 even when trained and tested only on increments the model refuses.

C.4 What this does and does not show

- **The sufficiency readout is not the planned output.** If the probes were reading "about to output INSUFFICIENT", they would fall to chance once the output is held fixed. They stay at 0.88–0.99 depending on the task. The residual stream at the answer-start position encodes that the context contains the complete evidence, separately from the decision to answer.

- **The internal signal is a better sufficiency detector than the model's self-report.** For a system that has to decide whether retrieved evidence is complete, the anchor activation, read before generation, is more accurate than asking the model and reading its answer: 0.98 against 0.71–0.77 AUROC with paragraph count matched. At 0.95 precision the probe accepts 86–91% of sufficient contexts with 4% false alarms, whereas the model's answers accept 63–71% with 15–21% false alarms.

- **This is a statement about evidence completeness, not about the answer.** The probe is trained on construction labels (all gold hops present). A high score on a wrongly rejected context says that the representation registers the evidence as complete. It does not say that the model could compute the answer from it. Whether it will is what uptake measures (Section 5), and uptake is lower and nearly orthogonal to sufficiency. Among the minimal sufficient contexts it rejects, the model may lack the bridging step, and the internal sufficiency state cannot supply it. Consistently, using the sufficiency score to stop retrieval does not by itself improve accuracy (Section O): stopping at a sufficient context the model will not answer from still yields INSUFFICIENT.

- **The causal picture is consistent with this division.** The one causally active direction, v_{dm}, acts as an abstention gate (Section 6): pushing along it makes the model answer, including on some sufficient contexts it was rejecting (positive steering adds 3 points of correct answers on Qwen and 18 on Llama), but also on distractors and false evidence. The sufficiency *readout* is not that gate: projecting the probe direction out changes nothing. The model represents completeness accurately along directions it does not use for the answer/abstain decision, and its decision is driven by a lower-dimensional, noisier gate.

- **Scope.** The comparison uses MuSiQue two-hop questions the models answer incorrectly closed-book, under the evidence-only instruc-

tion. With a more permissive instruction (Section F), the models abstain less on sufficient evidence and follow false evidence more, so the verbalised false-negative rate depends on the prompt; the internal update readout barely changes (0.945 against 0.948).

D State versus update: which representation carries the readout?

The probes of Table 1 act on the paired update $\Delta h = h(C_k) - h(C_{k-1})$. A natural question is how much of their accuracy comes from the *change* of state, and how much from information already present in either absolute state. If $h(C_k)$ alone did as well, the finding would be that the model's current state represents whether its evidence is sufficient. If $h(C_k)$ were much worse than Δh, sufficiency would be encoded specifically in the transition caused by the new evidence.

Protocol. For every update task we refit the probe on four representations of the *same* increments, with the same labels, document split, regularisation ($C = 0.5$, balanced classes) and layer selection (grouped three-fold cross-validation on training questions, over all blocks):

- $h(C_{k-1})$, the anchor state before the new evidence;
- $h(C_k)$, the anchor state after it;
- Δh, the paper's representation (it reproduces Table 1 to within 0.002);
- the concatenation $[h(C_{k-1}); h(C_k)]$, from which a linear probe can represent Δh, either state, or any other linear combination of the two.

Both states are the stored activations of the two stateless forward passes; Δh is their difference to within float16 precision.

Sufficiency. The state *after* the new evidence carries most of the readout. $h(C_k)$ reaches 0.930–0.940 on the full task and 0.900–0.908 on the matched one, against 0.947–0.952 and 0.910–0.917 for Δh. The state *before* the evidence is uninformative beyond the trajectory stage. On the full task it scores 0.733 in all three models, which is exactly the AUROC of an oracle that knows only whether the increment starts from the empty context, a partial context or an already sufficient one: 20% of increments starting from a partial context are decisive, and none of the others. On the matched task, where the decisive increment and its four controls start from the same state, it scores exactly 0.500. The concatenation matches Δh (0.949–0.950; matched 0.907–0.923). So subtracting the earlier state adds no information that having both states does not; what it does is remove the large question-specific component that both states share, which makes a single linear direction work.

Why $h(C_k)$ is not trivially enough. On the full task, 27% of negatives end in a context that is already sufficient: a distractor, duplicate or falsified paragraph added after sufficiency, or the full context. A pure "is this context sufficient?" detector cannot reject them. Applying the state-sufficiency probe of Table 1 (trained on whole contexts, 0.99 AUROC on its own task) to $h(C_k)$ of these increments gives only 0.70 on the full update task, and 0.87–0.90 on the matched task, where no negative ends in a sufficient context. The $h(C_k)$ probe can, because the state after the decisive paragraph differs from the state after a post-sufficiency addition: the former is a *minimal* sufficient context, the latter has more paragraphs. Part of what separates them is plausibly context size. On the matched task this cannot help, because every control adds exactly one paragraph to the same penultimate state. There, $h(C_k)$ at 0.90 is a genuine readout of sufficiency from the resulting state.

Uptake, correction, stability, revision. For the behavioural tasks the state after the evidence is as informative as the update or more so: uptake 0.769–0.798 against 0.771–0.783, correction 0.766–0.872 against 0.730–0.818, stability 0.813–0.882 against 0.750–0.869, revision 0.789–0.806 against 0.709–0.767. The concatenation is best or tied on most of them. The earlier state carries a little information of its own here (0.52–0.65): how the model behaves after new evidence depends partly on the state it was in. These tasks are about the model's *resulting* answer, so it is unsurprising that the resulting state predicts them best. The uptake result means the signal that decisive evidence will be integrated is present in the state after the evidence arrives, before generation, and does not require comparing with the state before it.

What this means for the paper's claims.

- The informational claims stand: sufficiency

Model	Task	$h(C_{k-1})$	$h(C_k)$	Δh	$[h(C_{k-1}); h(C_k)]$
Qwen2.5-7B	Sufficiency	0.733	0.930	0.947	**0.949**
	Sufficiency (matched)	0.500	0.901	0.911	**0.919**
	Uptake	0.605	**0.798**	0.775	0.780
	Correction	0.574	0.766	0.730	**0.782**
	Stability	0.549	0.813	0.750	**0.832**
	Revision	0.556	**0.806**	0.767	0.791
Llama-3.1-8B	Sufficiency	0.733	0.936	**0.952**	0.950
	Sufficiency (matched)	0.500	0.908	0.917	**0.923**
	Uptake	0.570	**0.794**	0.783	0.788
	Correction	0.522	0.872	0.818	**0.883**
	Stability	0.592	0.882	0.869	**0.888**
	Revision	0.563	**0.795**	0.709	0.751
Mistral-7B	Sufficiency	0.733	0.940	0.948	**0.949**
	Sufficiency (matched)	0.500	0.900	**0.910**	0.907
	Uptake	0.647	0.769	0.771	**0.775**
	Correction	0.566	**0.822**	0.790	0.819
	Stability	0.524	**0.857**	0.800	0.857
	Revision	0.634	0.789	0.721	**0.790**

Table 9: Held-out AUROC (document split, best layer by inner cross-validation) of probes for each update task, fitted on four representations of the same increments. Bold: best per row.

and uptake are linearly readable at the answer-start position before generation, robustly across splits and models. But "readable from the update" should be understood as "readable from the resulting state, measured in a design that controls for the starting state". The paired update is the cleanest way to read the state change without question-specific content, and it is what makes the matched controls possible (same starting state). It is not the source of the information.

- The causal analysis of Section 6 is unaffected. It patches the anchor residual of the insufficient prompt with that of the sufficient prompt, which is the same as adding Δh to $h(C_{k-1})$. Its conclusion, that the probe direction is inert while the full residual and v_{dm} are causal, concerns directions in this space whichever way they were found.

- The text-baseline comparison (Section 5.2) is unaffected: it compares classifiers on the same increments and labels.

E Robustness across data splits

Every probe is trained on 70% of questions and tested on the remaining 30% under four groupings that are never shared between train and test: question id; gold document, where questions that share any supporting page are grouped together (transitively) and each group goes entirely to one side; answer entity; and the tuple of relation types along the question's hops. The document split is the default in the main text. Table 10 gives the best-layer test AUROC under all four for the three two-hop runs, the 3–4-hop run and 2WikiMultiHopQA. Sufficiency varies by at most 0.015 across splits. Uptake is lower on the document and answer splits than on the question split, by 0.06–0.13 on the two-hop runs and by 0.18 on 3–4-hop questions (Section I).

F Evidence-augmented regime

Setup. The evidence-augmented regime repeats every stateless forward pass of the Qwen two-hop run with one change to the instruction: "use the supplied evidence together with your own knowledge" instead of "use only the supplied evidence". Conditions, increments, splits and probes are otherwise identical.

Results. The Qwen probes give sufficiency 0.945, matched 0.913, uptake 0.721 and correction 0.786 (document split), within 0.05 of the evidence-only regime. Behaviour shifts as expected for a more permissive instruction: on minimally sufficient contexts the model answers more (0.51 correct, 0.23 abstain) and it follows false evidence more often.

G Cross-task and zero-retraining transfer

Cross-task transfer. For every pair of tasks we apply the direction learned for one task (row) to the

Run	Task	question	document	answer	relation
Qwen2.5-7B	Suff.	0.956	0.948	0.945	0.947
Qwen2.5-7B	Uptake	0.851	0.775	0.726	0.841
Llama-3.1-8B	Suff.	0.960	0.952	0.958	0.959
Llama-3.1-8B	Uptake	0.845	0.783	0.736	0.815
Mistral-7B	Suff.	0.964	0.949	0.954	0.958
Mistral-7B	Uptake	0.873	0.771	0.758	0.793
Qwen 3–4 hop	Suff.	0.931	0.925	0.923	0.932
Qwen 3–4 hop	Uptake	0.795	0.613	0.611	0.795
Qwen 2Wiki	Suff.	0.986	0.986	0.981	0.983
Qwen 2Wiki	Uptake	0.894	0.894	0.866	0.786

Table 10: Best-layer AUROC under the four disjoint splits, for the three 2-hop runs, the 3–4-hop run and 2WikiMultiHopQA.



Figure 5: Cross-task transfer (Qwen2.5-7B, document split): AUROC of the direction learned for the row task, evaluated on the column task.

increments of another (column), at the row task's best layer, on the document split's test questions (Fig. 5). Uptake and correction share a direction (cross-task AUROC 0.82), whereas the sufficiency direction reaches only 0.54–0.60 on uptake.

Zero-retraining transfer. Table 11 applies the Qwen two-hop stateless directions, unchanged and at the source's best layer, to the conversational protocol (Section H), to 2WikiMultiHopQA (Section J) and to 3–4-hop MuSiQue (Section I). All target increments are scored, since none of them were used to train the source probe.

H Conversational protocol

Setup. In the conversational revision protocol the model sees, in a two-turn chat, the question with its earlier evidence and its own stateless answer A_k from that context, and then only the new paragraph, with the instruction to change its answer only if the total evidence warrants it. The paired update is $h_{\text{conv}}(C_k \to C_{k+1}) - h(C_k)$. The 10 chain-related increment kinds are re-run in this format (30,000 increments for Qwen); labels are defined as in the stateless protocol.

Results. Probes trained within the conversational regime reach sufficiency 0.951, uptake 0.818, correction 0.922, stability 0.959 and revision 0.969 (document split). Stability and revision are much higher than in the stateless regime because the model's previous answer is in the prompt. The stateless directions transfer at 0.79 (sufficiency), 0.78 (uptake), 0.92 (correction) and 0.82 (stability) (Table 11).

I 3–4-hop questions

Setup. A separate Qwen run uses 3- and 4-hop MuSiQue training questions (closed-book accuracy 1.3%), with the condition set of Section A including random hop orders and leave-one-out contexts, which exist only for $n \geq 3$.

Results. Sufficiency probes reach 0.931/0.925/0.923/0.932 (question/document/answer/relation split); matched 0.907/0.900/0.896/0.906; uptake 0.795/0.613/0.611/0.795. Along the sufficiency direction, chain-decisive and leave-one-out-decisive increments have the same mean projection (+0.023), followed by false (+0.003), distractor (−0.02), redundant and answer-string (−0.06) and non-decisive gold increments (−0.11). Patching (correct/answered): none 0.05/0.20, full 0.25/0.62, parallel 0.06/0.19, orthogonal 0.25/0.62, PC256 0.20/0.51, random-256 0.06/0.20; projecting out v_{dm} leaves 0.03 answered. The two-hop directions transfer at 0.82–0.85 for sufficiency (Table 11).

Direction (MuSiQue 2-hop, stateless)	→ conversational	→ 2WikiMultiHopQA	→ MuSiQue 3–4 hop
sufficiency	0.790	0.956	0.822
sufficiency matched	0.776	0.931	0.849
uptake	0.776	0.791	0.675
correction	0.920	0.846	0.744
stability	0.820	0.822	0.755
revision	0.659	0.770	0.708

Table 11: Zero-retraining transfer of Qwen2.5-7B 2-hop directions (AUROC at the source's best layer).

J 2WikiMultiHopQA

Setup. 2WikiMultiHopQA (Ho et al., 2020) was built from Wikipedia and Wikidata to evaluate multi-hop reasoning with explicit reasoning paths: each question comes with the evidence triples that connect it to the answer. Because it is constructed independently of MuSiQue, we use it to test whether the probes transfer to a different construction. The Qwen pipeline is run unchanged on its two-hop questions, with gold paragraphs from its supporting facts and its own distractors.

Results. Within-dataset probes (document split): sufficiency 0.986, matched 0.972, uptake 0.894, correction 0.914. Here the matched task is almost fully explained by nuisance covariates (0.965, chiefly the change in gold-answer log-probability); the residualised probe reaches 0.82. The MuSiQue directions transfer at 0.96 (sufficiency), 0.93 (matched), 0.85 (correction) and 0.79 (uptake) (Table 11).

K Closed-book transfer

Setup. This experiment asks whether the directions learned on evidence trajectories also say something when no evidence is given at all. We re-run the closed-book prompt on Qwen's screened pool of 4,000 questions with activation capture and score every stored direction on the closed-book anchor activations, for two targets: closed-book correctness and closed-book abstention (INSUFFICIENT). A probe trained directly on closed-book activations (question-disjoint split) gives the ceiling.

Results. Closed-book answers are correct 1.5% of the time and abstain 91.7% of the time. For correctness, the state-correctness direction reaches 0.810 (0.838 at its best layer), correction 0.799 and state-sufficiency 0.722 (0.808), against an own-probe ceiling of 0.874. The abstention-flag direction predicts abstention at 0.977, against a ceiling of 0.973.

L Steering along the logistic directions

Setup. We add $\alpha\, u\, v$ along the logistic state-sufficiency direction at its best layer, from the anchor onward, with u the sufficient-minus-insufficient gap along v (Section B). Doses are $\alpha \in \{\pm 0.5, \pm 1, \pm 2\}$ gap units ($\leq$ 1.4 ℓ_2) and $\alpha \in \{\pm 8, \pm 16, 32\}$ (5–22 ℓ_2, on a mean residual norm of 61). We also steer along the uptake direction at layer 26 ($\leq$ 16 ℓ_2), and run an all-positions variant. A random unit direction with the same ℓ_2 magnitude is the control. Outcomes are measured on 150 held-out questions in each of five sets: sufficient, insufficient, false-evidence, answer-string and distractor-only contexts.

Results. Every rate stays within ± 0.02 of baseline on all five sets, for the probe directions and the random control alike. At 32 units all rates degrade by 0.02–0.05, again equally for probe and random directions. The all-positions variant at $\leq$ 11 ℓ_2 is equally flat.

M Difference-of-means steering

Low dose (main protocol). We add $\alpha\, u_l\, v_{\mathrm{dm}}^{(l)}$ at four layers from the best sufficiency layer, from the anchor onward, with $\alpha \in \{\pm 0.5, \pm 1\}$ in units of the per-layer sufficient-minus-insufficient gap, on 150 held-out questions per set. Table 12 gives the full numbers behind Fig. 4.

High dose. Adding $\alpha\, u_l\, v_{\mathrm{dm}}^{(l)}$ at all positions and at every layer from the best layer onward with $|\alpha| \geq 2$ (per-layer gaps of 5–14 ℓ_2) produces degenerate output: every prompt is "answered", never correctly and never with INSUFFICIENT. This happens equally for the probe direction and the random control, and is why the main protocol uses four layers, anchor onward and $|\alpha| \leq 1$.

N Single-layer patching

Setup. As in Section 6, the insufficient prompt's anchor residual is replaced by the sufficient run's

Model	α	sufficient corr./ans.	insufficient ans.	distractors ans.	false ans./followed	answer-string ans.
Qwen2.5-7B	-1.0	0.23/0.34	0.05	0.00	0.23/0.06	0.03
	-0.5	0.34/0.52	0.09	0.01	0.43/0.13	0.09
	$+0.5$	0.49/0.79	0.20	0.06	0.75/0.25	0.29
	$+1.0$	0.49/0.82	0.26	0.12	0.79/0.25	0.35
Llama-3.1-8B	-1.0	0.03/0.07	0.00	0.00	0.03/0.01	0.00
	-0.5	0.25/0.38	0.07	0.00	0.23/0.12	0.05
	$+0.5$	0.65/0.89	0.27	0.13	0.82/0.35	0.34
	$+1.0$	0.69/0.94	0.51	0.29	0.92/0.39	0.59
Mistral-7B	-1.0	0.01/0.08	0.06	0.11	0.06/0.01	0.04
	-0.5	0.11/0.33	0.09	0.01	0.25/0.09	0.04
	$+0.5$	0.47/0.78	0.31	0.12	0.75/0.29	0.29
	$+1.0$	0.54/0.87	0.47	0.28	0.87/0.28	0.43

Table 12: Low-dose steering along v_{dm} (150 held-out questions per set). Baselines ($\alpha = 0$) appear in Fig. 4.

(full patch), but at a single layer only instead of every layer from the best one onward.

Results. Patching only the best sufficiency layer (16 for Qwen) leaves the insufficient prompt unchanged in all variants. Patching only layer 26 with the full sufficient residual restores $0.18/0.62$ (correct/answered), so late-layer anchor content alone suffices to make the model answer.

O Adaptive retrieval

The readouts of Section 5 are computed before generation, so they can drive retrieval: after each round, read the anchor and decide whether to stop and answer or to retrieve more. We test every learned direction in this role, both as a *stopping rule* and as a *verifier* of the answer given at the stopping round.

Simulation. Each held-out question is walked along its canonical trajectory, one retrieval round per stage: (1) the matched distractors, (2) the first gold hop, (3) the minimal sufficient context, (4) the sufficient context plus a distractor. A policy that never stops answers from stage 4; an oracle stops at stage 3. At round k a policy reads either the state $h(C_k)$ (state probes) or the update the round produced, $\Delta h = h(C_k) - h(C_{\mathrm{prev}})$ (update probes), projects it on the direction at the direction's best layer, and stops when the projection crosses a threshold. The threshold *and the sign* are chosen on training questions to maximise accuracy at the stopping stage (ties broken by fewer rounds), so each direction is used in whichever reading serves retrieval best. For stability and revision we also report the reading that stops when the probe says the last round *changed the answer*. Baselines: never stop; the oracle; the behavioural rule, which stops at the first round whose greedy answer is not INSUFFICIENT (it needs a generation per round, which the probes do not); and a next-token-entropy threshold calibrated the same way. Accuracy counts abstentions as wrong. Everything is computed from stored activations and behaviour, with no new forward passes. Thresholds are calibrated on the questions the probes were trained on, which makes the probe policies, if anything, slightly over-confident on test questions. The trajectories present gold hops in order, so the simulation measures *when to stop*, not what to retrieve.

Stopping rules (Table 13, Fig. 6). Only the sufficiency signals are useful stopping rules. The update-sufficiency v_{dm} direction is the best learned rule on all three MuSiQue models (0.478, 0.468, 0.336). It matches the oracle (0.475, 0.470, 0.336) in about three rounds, and on Qwen it is the one rule significantly better than the behavioural rule. The logistic state and update sufficiency probes are statistically tied with the behavioural rule on MuSiQue, while acting before any generation. On HotpotQA both beat it (0.732 and 0.728 vs 0.650), and on the synthetic data the state probe does (0.977 vs 0.939), because the behavioural rule stops too early there (HotpotQA: 2.53 rounds). On 2WikiMultiHopQA they tie it (0.583 vs 0.593). The other directions are weak stopping rules on every source. Uptake, correction and stability are trained on decisive or already-correct increments and say nothing about whether the evidence is complete. Their best thresholds stay within 0.06 of the never-stop policy on MuSiQue (0.24–0.42) and below the best sufficiency rule on every source. Read as "stop when the answer changes", stability never fires (about 4.0 rounds, at full-context accuracy). Revision, the direct change detector, stops at about three rounds but often on the first wrong answer, and is the worst learned rule on

Stopping rule	MuSiQue Qwen	MuSiQue Llama	MuSiQue Mistral	HotpotQA Qwen	2WikiMultiHopQA Qwen	Synthetic Qwen
Baselines						
full context (never stop)	0.404 (4.00)	0.385 (4.00)	0.243 (4.00)	0.685 (3.99)$^{+}$	0.460 (4.00)	0.966 (3.65)$^{+}$
oracle (stop when sufficient)	0.475 (3.00)	0.470 (3.00)	0.336 (3.00)	0.740 (3.00)$^{+}$	0.583 (3.00)	0.973 (2.93)$^{+}$
behavioural (first non-INSUFFICIENT)	0.460 (3.09)	0.460 (3.09)	0.318 (3.08)	0.650 (2.53)	**0.593 (3.10)**	0.939 (2.88)
next-token entropy	0.381 (3.85)$^{-}$	0.378 (3.96)	0.261 (3.91)	0.690 (3.74)$^{+}$	0.467 (3.67)$^{-}$	0.969 (3.35)$^{+}$
State probes, $h(C_k)$						
sufficiency (logistic)	0.466 (3.13)	0.448 (3.11)	0.326 (3.13)	**0.732 (3.08)**$^{+}$	0.583 (3.02)	**0.977 (2.96)**$^{+}$
sufficiency (v_{dm})	0.407 (3.94)	0.385 (3.58)	0.261 (3.63)	0.690 (3.86)$^{+}$	0.463 (3.71)	0.967 (3.00)$^{+}$
correctness	0.399 (3.57)	0.443 (3.51)	0.284 (3.64)	0.667 (2.92)	0.547 (3.42)	0.964 (3.23)$^{+}$
abstention flag	0.435 (3.07)$^{-}$	0.435 (3.12)$^{-}$	0.302 (3.13)$^{-}$	0.702 (3.18)$^{-+}$	0.583 (3.12)$^{-}$	0.969 (2.99)$^{-+}$
Update probes, Δh of the round						
sufficiency (logistic)	0.458 (3.30)	0.460 (3.09)	0.316 (3.18)	0.728 (3.23)$^{+}$	0.570 (3.16)	0.949 (3.32)
sufficiency (v_{dm})	**0.478 (3.06)**$^{+}$	**0.468 (2.98)**	**0.336 (3.00)**	0.727 (3.07)$^{+}$	0.543 (3.32)	0.970 (3.15)$^{+}$
sufficiency matched	0.404 (4.00)	0.385 (4.00)$^{-}$	0.243 (4.00)	0.683 (3.98)$^{+}$	0.513 (3.21)	0.858 (3.22)
uptake	0.417 (3.55)	0.388 (3.35)	0.259 (3.47)	0.683 (3.98)	0.507 (3.42)	0.957 (3.07)$^{+}$
correction	0.404 (3.43)	0.405 (3.38)	0.302 (3.53)	0.667 (2.93)	0.550 (3.36)	0.920 (3.19)
stability	0.407 (3.46)	0.400 (3.42)	0.243 (4.00)$^{-}$	0.685 (3.99)$^{+}$	0.543 (3.38)	0.964 (3.63)$^{+}$
stability, stop when the answer changes	0.402 (3.99)	0.385 (4.00)	0.243 (4.00)	0.685 (3.99)	0.457 (3.99)	0.966 (3.65)
revision (stop when the answer changes)	0.396 (3.13)	0.363 (3.12)	0.236 (3.23)	0.690 (3.90)$^{+}$	0.513 (3.07)	0.934 (3.02)
test questions	609	600	610	600	300	830

Table 13: Every direction as a stopping rule: accuracy at the stopping round (mean rounds in parentheses), held-out questions of the document split. Bold: best non-oracle rule per column. $^{-}$: the training-optimal reading stops when the projection is *low*. $^{+}$: the 95% paired bootstrap interval of the accuracy difference to the behavioural rule lies above zero (1,000 resamples of test questions).

MuSiQue (0.236–0.396). The v_{dm} direction of the *state*, which the causal analysis identified as an abstention gate (Section 6), is a poor stopping rule (0.26–0.41 on MuSiQue). Its update counterpart is the best one. Figure 6 sweeps each threshold on test questions: the sufficiency curves peak at the oracle's operating point, while the uptake and stability curves rise roughly monotonically towards the full-context accuracy and never cross the behavioural rule on MuSiQue.

Verification at the stopping round (Table 14). The directions that fail as stopping rules are useful once retrieval has stopped. We take the stopping round chosen by the state-sufficiency rule and ask, among the questions the model answers there, how well each signal ranks correct above wrong answers. This is the decision to output, flag or re-retrieve. The uptake direction is the best learned verifier on MuSiQue Qwen (0.672) and 2WikiMultiHopQA (0.847), where it beats output confidence (top-token probability 0.621, 0.705). Correction is the best learned verifier on Mistral (0.666 vs 0.630). By AUROC, next-token entropy is ahead on Llama and HotpotQA (by 0.02) and on the synthetic data (by 0.11, where 99% of answered stops are correct). For selective answering, keeping only the more confident half of the answers, uptake gives higher accuracy than top-token probability in five of six settings (0.789 vs 0.751 on MuSiQue Qwen, 0.957 vs 0.870 on 2WikiMultiHopQA), and is level on the synthetic data (0.995 vs 0.998). The sufficiency directions are near chance as verifiers on MuSiQue and HotpotQA (0.46–0.60), and the abstention flag is below chance everywhere. They detect completeness, which the stopping rule has already selected for, not whether the answer is right. This division of labour mirrors the readout results. Sufficiency tells the system when the evidence is complete. Uptake, nearly orthogonal to it (Section 5), tells whether the model has actually integrated that evidence into its answer.

Other sources. On 3–4-hop MuSiQue the trajectories end at the sufficient context (no post-sufficiency stage), so the oracle and the never-stop policy coincide (0.360). All learned rules reach this value, while the behavioural rule stops too early (0.340). On RAGBench (relation split, 955 test questions) most trajectories have two or three stages and no distractor after sufficiency, and every policy lies within 0.21–0.23. With the original protocol of a threshold at 0.97 training precision, the state-sufficiency rule reached 0.442 in 2.87 rounds on MuSiQue Qwen and 0.707 in 2.90 on HotpotQA. Accuracy-optimal calibration trades a few rounds for accuracy on MuSiQue.

P Results by question group on the external sources

Table 15 breaks down the held-out probe AUROC of the external sources (Section 7) by question group, to separate the evidence-state readout from properties of the questions. The groups are: closed-

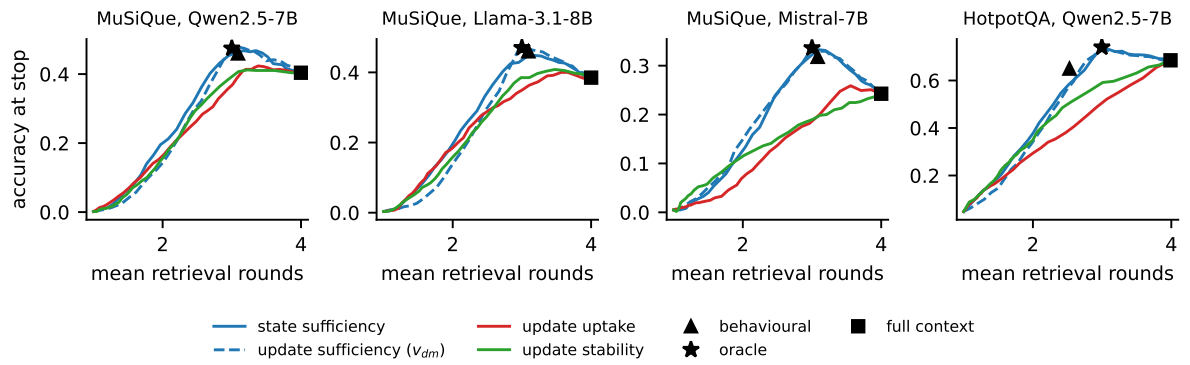


Figure 6: Accuracy against mean retrieval rounds when each direction's stopping threshold is swept over test questions (41 quantiles), with the behavioural rule, oracle and full context as points.

Signal at the stopping round	MuSiQue Qwen	MuSiQue Llama	MuSiQue Mistral	HotpotQA Qwen	2WikiMultiHopQA Qwen	Synthetic Qwen
top-token probability	0.621	0.615	0.630	0.654	0.705	0.876
−next-token entropy	0.623	**0.630**	0.637	**0.657**	0.705	**0.877**
state: sufficiency	0.465	0.463	0.521	0.526	0.621	0.738
state: correctness	0.560	0.568	0.618	0.626	0.763	0.388
state: abstention flag	0.432	0.473	0.412	0.395	0.444	0.252
update: sufficiency	0.482	0.543	0.495	0.554	0.594	0.585
update: sufficiency (v_{dm})	0.566	0.569	0.601	0.567	0.806	0.534
update: uptake	**0.672**	0.613	0.647	0.617	**0.847**	0.764
update: correction	0.594	0.576	**0.666**	0.639	0.753	0.400
update: stability	0.549	0.566	0.559	0.562	0.775	0.497
update: revision	0.512	0.517	0.490	0.524	0.601	0.531
accuracy of all answered stops	0.665	0.666	0.534	0.810	0.761	0.990
accuracy, top half by top-token prob.	0.751	0.733	0.618	0.863	0.870	0.998
accuracy, top half by uptake	0.789	0.757	0.651	0.867	0.957	0.995
answered stops	427	404	373	542	230	819

Table 14: Verification once retrieval stops (state-sufficiency stopping rule): AUROC of each signal at the stopping round for the correctness of the answer given there (answered stops only). Bold: best per column. Bottom: accuracy of all answered stops and of the half ranked highest by top-token probability or by the uptake direction.

book status (unlike the main runs, HotpotQA and RAGBench keep the questions the model answers correctly without evidence); question type (HotpotQA bridge versus comparison, whose two facts are independent lookups); domain (RAGBench's twelve sub-datasets grouped into biomedical, finance, technical, customer support, general web QA and Wikipedia); and hop count (synthetic 1–3 hops; for RAGBench, one versus two gold documents). The stored best-layer probe of each source, trained on all groups, is scored per group on test questions without refitting; increment counts are in parentheses.

Q The Evidence State Transition Benchmark

The Evidence State Transition Benchmark (ESTB) tests whether the evidence-state readout and its causal profile, established on MuSiQue trajectories in Sections 5 and 6, survive controlled changes to the evidence. It puts all of our sources under one labelling scheme and one evaluation protocol. This appendix describes what the benchmark measures, how each axis was built, and the full results of the retrieval-source and linguistic-form runs, which appear only in summary in Section 7.

Q.1 What the benchmark measures

Unit of measurement. As in the rest of the paper, the unit is an *evidence state transition*: an increment $C_{k-1} \to C_k$ that adds one paragraph (or, for post-sufficiency and full-context increments, a set of paragraphs) to the context, with the paired update $\Delta h = h(C_k) - h(C_{k-1})$ read at the answer-start anchor. ESTB annotates every increment with the *quality state* of the evidence it adds (Table 16) and with the values of the other axes (Table 17): retrieval source, linguistic form, domain, question type, hop count and closed-book status.

Labels. Each increment carries the sufficiency label $y^{\mathrm{suff}} = \mathbf{1}[C_k$ sufficient, C_{k-1} not], which is defined by construction from the gold hops and does not depend on the model, and a behavioural uptake label for increments that cross sufficiency

Dataset	grouping	group	sufficiency	uptake	correction
HotpotQA	closed book	incorrect	0.978 (7840)	0.724 (986)	0.839 (5554)
HotpotQA	closed book	correct	0.984 (1689)	0.802 (214)	0.885 (760)
HotpotQA	qtype	bridge	0.978 (7878)	0.680 (982)	0.839 (4963)
HotpotQA	qtype	comparison	0.985 (1651)	0.787 (218)	0.880 (1351)
RAGBench	domain	biomedical	0.929 (1473)	0.776 (291)	0.818 (1331)
RAGBench	domain	finance	0.899 (1081)	0.641 (206)	0.769 (949)
RAGBench	domain	technical	0.881 (1069)	0.675 (196)	0.733 (989)
RAGBench	domain	customer support	0.897 (716)	0.569 (153)	0.658 (587)
RAGBench	domain	general	0.835 (1821)	0.721 (352)	0.702 (1520)
RAGBench	domain	wikipedia	0.978 (729)	0.557 (158)	0.766 (612)
RAGBench	n hops	1	0.836 (2476)	0.698 (444)	0.785 (2211)
RAGBench	n hops	2	0.931 (4413)	0.677 (912)	0.711 (3777)
Synthetic	n hops	1	0.922 (3645)	0.992 (486)	0.875 (2450)
Synthetic	n hops	2	0.970 (6192)	0.928 (1032)	0.926 (4929)
Synthetic	n hops	3	0.970 (11501)	0.916 (2395)	0.931 (8859)

Table 15: Held-out probe AUROC by question group (test-set increment counts in parentheses): closed-book status, question type, domain and hop count.

(answer becomes correct or not). An increment crosses sufficiency exactly when it supplies the last missing hop, whether in the original wording (C3), paraphrased, or from an independently authored source.

Measurements per cell. A *cell* is the set of increments of one run that share one value on one axis (for example, "E5-retrieved distractors, quality state C1"). For every cell we report:

1. **Within-run readout:** the test AUROC of the run's own held-out probe (trained on the run's training questions), for sufficiency, matched sufficiency and uptake. This is the ceiling: is the state linearly decodable at all under this kind of evidence?

2. **Transfer without retraining:** the AUROC of the Qwen MuSiQue two-hop direction (document split, at its best layer), applied unchanged. Because the ESTB runs sample from the same MuSiQue pool as the source run, 818 of the 1,200 questions in the benchmark-gold runs (273 of 402 and 152 of 219 in the fully retrieved runs) are in the source probe's training split. These questions are *excluded* from every transfer score, so transfer is measured only on questions the source probe never saw.

3. **Quality-state separability:** for each quality state Q, the AUROC separating Q's increments (negatives) from the decisive increments of the same run (positives). Near 1, the readout cleanly distinguishes Q from completing the evidence; lower values mean that Q looks like sufficiency to the probe.

4. **Causal profile:** on 150 held-out questions, the answered rate of the insufficient prompt unpatched, with the full sufficient anchor residual patched in, and with only the probe-direction component ($\| \ v$) patched in, all at every layer from the best one onward; and the answered rate of sufficient prompts after projecting v_{dm} to its insufficient-state value (Section 6).

The benchmark therefore separates three questions: whether the state is *present* (1), whether it is the *same* state as on MuSiQue (2, 3), and whether its causal profile, an inert probe direction alongside a causally sufficient anchor and a v_{dm} abstention gate, carries over (4).

What the benchmark does not measure. ESTB does not score answer quality or retrieval quality for their own sake: retrieval and paraphrasing only vary the evidence that enters a trajectory. It does not ask whether the model *should* answer from partial or contradictory evidence; those states are scored by whether the readout separates them from genuine completion. The sufficiency label is structural (all hops present) rather than semantic, so a passage that happens to state a fact in a way our string-based labelling misses (Section Q.7) is counted as insufficient.

Q.2 Evidence quality states

Table 16 lists the seven quality states of the evidence-quality axis and the increment kinds (Section A) that realise them. All states except C0 are increments, i.e. transitions *into* the state from the preceding context.

The contradictory state has two sources. On benchmark and retrieved trajectories, the false value is another question's sub-answer of the same relation type (`false_synthetic`); on the synthetic

State	Name	Increment kinds (added evidence)	y^{suff}
C0	none	no evidence (start state of every trajectory)	–
C1	irrelevant	`distractor_from_none`, `distractor`: a length-matched distractor added to an empty or penultimate context	0
C2	partial	`gold_nondecisive`: a gold hop that leaves another hop missing; `partial`: the decisive paragraph rewritten to withhold its value; `redundant` from a penultimate state (C2+redundant)	0
C3	complete	`decisive`, `loo_decisive`: the last missing hop; `decisive_paraphrase`, `decisive_independent`: the same hop paraphrased or from an independent source	1
C4	complete + redundant	`post_distractor`, `post_redundant`, `post_redundant_paraphrase`, `post_redundant_independent`, `full_context`: further evidence added after sufficiency	0
C5	contradictory	`false`: the decisive hop with its value replaced (from a penultimate state); `post_false`: a falsified copy added to a sufficient context	0
C6	answer string, no relation	`answer_ctrl`: a distractor with "(Not to be confused with {answer}.)" appended	0

Table 16: ESTB evidence-quality states and the increment kinds that realise them. y^{suff}: whether the increment crosses into sufficiency.

benchmark, it is a generated value of the same type. The tables label the first kind "C5 contradictory (synthetic)" because the false value is substituted rather than found in a real document; naturally occurring contradictions are not separated from `near_miss` passages by our labelling (Section Q.7). Partial evidence is defined in two ways: for multi-hop questions, a gold hop that leaves another hop missing ("partial (hop)"); for any question, the decisive paragraph rewritten so that its subject and relation remain but the value is withheld ("partial (masked)", Section Q.5).

Q.3 Axes and sources

All ESTB runs reported here use Qwen2.5-7B-Instruct (Qwen Team, 2024) and the pipeline of the main text without modification. The four retrieval runs and the variants run draw 1,200 two-hop MuSiQue training questions that the model answers incorrectly closed-book, with no random hop permutations.

Q.4 Retrieval-source axis

Corpus and retrievers. Evidence comes from wiki-18, the December 2018 Wikipedia dump of Karpukhin et al. (2020) as distributed with FlashRAG (Jin et al., 2025): 21M passages of 100 words with their article titles. *BM25* (Robertson and Zaragoza, 2009), a lexical retriever, uses the `bm25s` implementation (Lù, 2024) over title and text, with English stop-words and Snowball stemming. *E5* (Wang et al., 2022), a dense retriever, uses the published FAISS flat inner-product index of `e5-base-v2` passage embeddings; queries are encoded with the `query:` prefix and mean pooling, then ℓ_2-normalised.

Queries. A single question-level query rarely retrieves both hops of a bridge question. Each question therefore issues the question itself plus one query per hop, taken from MuSiQue's decomposition with the earlier hops' sub-answers substituted for the `#k` placeholders (three queries per two-hop question). This imitates an iterative or decomposing RAG system that gets the earlier hops right. The per-query hit lists are merged round-robin by rank, de-duplicated, and cut at $k = 30$ passages per question.

Passage labels. Each retrieved passage is labelled against the question's gold hops by title and string matching (titles normalised by lower-casing and removing a trailing parenthetical):

- `retrieved_gold`: the title matches a gold hop's page *and* the passage contains that hop's sub-answer;
- `same_page`: the title matches a gold hop's page but the sub-answer is absent, i.e. partial evidence found in the wild;
- `near_miss`: not a gold page, but the passage mentions a bridge entity or the final answer, which is the natural analogue of the answer-string control C6;
- `irrelevant`: everything else.

Table 18 gives the label composition. E5 retrieves at least one gold passage for 85.5% of questions and both hops for 33.5%; BM25 does so for 68.1% and 18.2%.

Trajectories. Two modes turn retrieval output into trajectories.

Axis	Levels	Source (runs)
Evidence quality	C0–C6 (Table 16)	every run
Retrieval source	benchmark; BM25; E5 (dense); each as distractors only or as all evidence	MuSiQue 2-hop rebuilt over wiki-18 (Section Q.4)
Linguistic form	verbatim; LLM paraphrase; independently authored; templated (5 families)	MuSiQue variants run (Section Q.5); synthetic benchmark
Domain	Wikipedia; biomedical; finance; technical; customer support; general web QA	MuSiQue, HotpotQA, 2WikiMultiHopQA; RAGBench's 12 sub-datasets
Question type	bridge; comparison; single-hop	HotpotQA; RAGBench; synthetic
Hop count	1; 2; 3–4	synthetic, RAGBench; MuSiQue 2-hop; MuSiQue 3–4-hop
Closed-book status	correct; incorrect	HotpotQA and RAGBench (`select: all`)

Table 17: ESTB axes. The domain, question-type, hop-count and closed-book axes reuse the external runs of Section 7, scored per group in Section P; the retrieval-source and linguistic-form axes required new construction and are detailed here.

	retrieved passages (top 30)				questions with gold for	
Retriever	gold	same page	near miss	irrelevant	≥ 1 hop	both hops
E5	0.080	0.139	0.292	0.489	0.855	0.335
BM25	0.053	0.080	0.331	0.535	0.681	0.182
Distractors used in the trajectories (18 per question): same page / near miss / irrelevant						
E5	0.231 / 0.460 / 0.309					
BM25	0.133 / 0.535 / 0.332					

Table 18: Retrieved passages by label (1,200 MuSiQue two-hop questions, multi-query retrieval, $k = 30$).

- **Retrieved distractors** (*noise* mode): the benchmark gold paragraphs are kept and every distractor is replaced by a retrieved non-gold passage. The 18 distractors per question are taken hardest first: same-page passages, then near misses, then irrelevant passages, each in rank order. The trajectory builder then draws length-matched distractors from this pool exactly as it does from the benchmark pool. Distractors are thus no longer the benchmark's curated negatives but what a retriever returns for the same question, and nearly a quarter of E5's distractors come from the gold pages themselves.

- **All evidence retrieved** (*full* mode): the gold paragraph of each hop is also replaced, by the highest-ranked `retrieved_gold` passage for that hop, so every paragraph the model sees is a 100-word wiki-18 passage. Only questions with a retrieved gold passage for both hops can be used: 402 for E5 and 219 for BM25.

Every other part of the construction (chains, controls, falsification, answer-string hatnotes, post-sufficiency and full-context increments) is unchanged, which yields the full set of quality states over retrieved evidence: 20,400 increments in each noise run, 6,809 (E5) and 3,703 (BM25) in the full runs.

Q.5 Linguistic-form axis

LLM paraphrase and partial variants. For every gold hop paragraph of the 1,200 questions, Qwen2.5-7B-Instruct generates two rewrites by greedy decoding (at most 220 tokens):

- *paraphrase*: "Rewrite the following encyclopedia paragraph in completely different wording and sentence structure, keeping every fact exactly the same. You must keep the exact string '{sub-answer}' verbatim."

- *partial*: "Rewrite the following encyclopedia paragraph so that it still talks about '{subject}' and the same topic, but withholds the specific fact '{sub-answer}': do not mention '{sub-answer}' or any equivalent name or value anywhere; instead say that the detail is not recorded here."

Generations are validated before use. A paraphrase is kept only if it contains the sub-answer, is between $0.5\times$ and $2\times$ the original's length, and shares fewer than 60% of its word 4-grams with the original. A partial is kept only if it contains neither the sub-answer, nor the final answer, nor any answer alias, has at least 15 words, and still mentions the hop's subject. Of 2,400 hop paragraphs, 2,180 (90.8%) yield a valid paraphrase and 1,411 (58.8%) a valid partial.

Independently authored sources. MuSiQue, HotpotQA and 2WikiMultiHopQA all take paragraphs from Wikipedia, but from different dumps, with different paragraph boundaries and often different revisions. For each MuSiQue gold hop with page title T and sub-answer a, we search the gold and distractor paragraphs of the *other* two datasets for one whose normalised title is T, that contains a, and that shares fewer than 60% of its 4-grams with the MuSiQue paragraph; when several qualify, the least overlapping is used. This finds an independently written statement of the same relation for 228 of the 2,400 hops (144 from HotpotQA, 84 from 2WikiMultiHopQA).

Increments. From the penultimate state, the paraphrased and independent paragraphs give `decisive_paraphrase` and `decisive_independent` increments (C3, crossing sufficiency), and the partial paragraph gives a `partial` increment (C2, which does not). After sufficiency, the paraphrase or independent version of a hop already present gives `post_redundant_paraphrase` and `post_redundant_independent` increments (C4). Together with the standard kinds, the variants run has 25,632 increments, including 2,180 paraphrased decisive, 228 independent decisive and 1,411 partial ones.

Synthetic templated form. The synthetic benchmark (Section 7) supplies the fully controlled end of the axis: fictional entities over seven relations (founder, birthplace, capital, spouse, award, employer, headquarters), five template families per relation, one to three hops, and for every gold fact a paraphrase, a partial (value withheld) and a contradictory (different value) variant, with near-miss distractors (same relation, other subject; same subject, other relation) and unrelated filler.

Q.6 Results

Retrieval source. With retrieved distractors (Table 19), within-run sufficiency is unchanged (0.946 E5, 0.951 BM25, against 0.948 with benchmark distractors), and the MuSiQue direction transfers at 0.935/0.939 on source-unseen questions. Retrieved distractors are harder than the benchmark's: the transferred direction separates C1 increments from decisive ones at 0.906 (E5) and 0.917 (BM25), against 0.945 for benchmark distractors, and answer-string controls at 0.962/0.971 against 0.982. The run's own probe recovers most of this (0.972/0.982 for C1), so the drop is a shift in the distractor distribution rather than a loss of the state. Uptake falls within run (0.68, against 0.78), but the transferred MuSiQue uptake direction scores 0.817 on the same increments: these runs have 1,200 rather than 2,000 questions, so the within-run probe is trained on fewer decisive increments, which is the likely reason. With all evidence retrieved, sufficiency stays at 0.91–0.92 within run and 0.905–0.910 transferred, and C1 separability is the lowest of any run (0.84 transferred, 0.89–0.92 own), which is expected when every paragraph is a 100-word passage from the same corpus as the gold ones. In every retrieval cell the causal profile of Section 6 reproduces: the full anchor patch raises answering from 0.10–0.30 to 0.68–0.77, the probe-direction component leaves it exactly at the unpatched rate, and projecting out v_{dm} leaves 0.01–0.03 answered.

Linguistic form. Paraphrased and independently authored decisive paragraphs are read as completing the evidence just as the original wording is. Against all non-decisive increments, the transferred MuSiQue direction scores original decisive increments at 0.931 AUROC, paraphrased ones at 0.929 and independently authored ones at 0.918, with mean projections 0.101, 0.098 and 0.096 (non-decisive: -0.082). Paraphrastic and independent redundancy after sufficiency is cleanly separated from completion (0.974 and 0.995 own probe, Table 20). The two hardest quality states are the ones that most resemble completion. Contradictory evidence added from the penultimate state (C5, the decisive paragraph with a swapped value) reaches only 0.76 own and 0.73 transferred, and it has the same "supplies the missing relation" surface form as the decisive paragraph. Masked partial evidence (C2), with subject and relation present but the value withheld, reaches 0.81 own and 0.74 transferred; its mean projection (0.044) lies between the non-decisive and decisive means. Partial evidence in the multi-hop sense (a non-decisive hop) is separated almost perfectly (1.00). Contradictory evidence is the hardest control in every retrieval run as well (0.71–0.75 transferred) and is also the state that projects closest to completion on the synthetic benchmark (Section 7).

Summary. Across the retrieval-source and linguistic-form axes, the sufficiency state remains decodable (≥ 0.91), the MuSiQue direction transfers to unseen questions (≥ 0.90), wording and

Evidence source	n_q	within-run probe (AUROC) suff.	matched	uptake	MuSiQue dir. suff.	uptake	answered rate unpatched	full	$\parallel v$	proj.-out v_{dm}
Benchmark gold + benchmark distractors	2000	0.948	0.911	0.775	(source)	(source)	0.13	0.71	0.13	0.01
Gold + E5-retrieved distractors	1200	0.946	0.909	0.681	0.935	0.817	0.10	0.77	0.10	0.02
Gold + BM25-retrieved distractors	1200	0.951	0.915	0.679	0.939	0.817	0.10	0.77	0.10	0.01
All evidence E5-retrieved	402	0.922	0.891	0.839	0.905	0.762	0.19	0.72	0.19	0.03
All evidence BM25-retrieved	219	0.912	0.876	0.748	0.910	0.820	0.30	0.68	0.30	0.02
+ LLM paraphrase / partial / independent	1200	0.935	0.885	0.718	0.929	0.796	0.10	0.77	0.10	0.00

Table 19: Retrieval-source and linguistic-form axes (Qwen2.5-7B, MuSiQue 2-hop). Within-run probes on the document split, except the two fully retrieved runs (question split, too few questions for document components). MuSiQue direction: transfer without retraining, scored only on questions outside the source probe's training split. Answered rate: the insufficient prompt unpatched, with the full sufficient anchor, and with only the probe-direction component; and the sufficient prompt with v_{dm} projected to its insufficient value.

Quality state vs. decisive increment	n	own probe	MuSiQue dir.
C5 contradictory (synthetic)	764	0.762	0.725
C2 partial (masked)	436	0.809	0.736
C1 irrelevant	1146	0.942	0.941
C4 complete+paraphrase	381	0.974	0.952
C6 answer string, no relation	764	0.975	0.979
C2+redundant	764	0.982	0.978
C4 complete+independent	57	0.995	0.950
C4 complete+irrelevant	382	0.997	0.980
C4 complete+redundant	382	0.999	0.984
C5 complete+contradictory	382	1.000	0.994
C2 partial (hop)	764	1.000	0.996
C4 full context	382	1.000	0.999

Table 20: Quality-state separability (MuSiQue variants run): AUROC separating each state's increments (negatives) from decisive increments (positives), with the run's own probe (test questions) and with the MuSiQue direction (source-unseen questions). Lower values mean the state looks more like completing the evidence.

authorship of the decisive evidence do not matter, and the causal dissociation reproduces in every cell. What changes is the difficulty of the controls: retrieved distractors from the gold pages and value-withheld or value-swapped versions of the decisive paragraph are where a deployed sufficiency detector would err.

Q.7 Limitations of the construction

- **Oracle decomposition.** The per-hop retrieval queries substitute gold sub-answers into MuSiQue's decomposition, so the retriever behaves like an iterative system that solves the earlier hops correctly. Gold coverage is an upper bound on what a realistic multi-hop retriever would achieve.
- **String-based passage labels.** `retrieved_gold` requires the gold page title and the sub-answer string. A passage on another page that states the same relation is labelled `near_miss` and used as a distractor; a gold-page passage that mentions the sub-answer in an unrelated sentence counts as gold. Both introduce some label noise, in opposite directions.
- **Selection in the fully retrieved runs.** Only questions whose two hops were both retrieved are kept (34% for E5, 18% for BM25), which favours questions with lexically salient hops.
- **Self-generated paraphrases.** Paraphrases and partials are generated by the same model that is probed. Validation removes rewrites that drop the sub-answer or copy the original, but generated text may still be easier for the model to read than human-written text. The independently authored paragraphs avoid this but are few (64 decisive increments on source-unseen questions).
- **One model.** All ESTB runs reported here use Qwen2.5-7B-Instruct.

R Compute and reproducibility

Collection for one 2-hop model ($\approx$ 35k conditions) takes about 22 hours on 64 CPU cores (bf16, $\approx$ 300

prompt tokens/s) or about one hour on an H200 shared with another job; the 3–4-hop run took 1.3 GPU-hours; all causal experiments and text baselines together used about 7 GPU-hours. Probing runs on CPU in one to two hours per model. Every stage is deterministic given the seed (question sampling, insertion positions, false-answer choice, splits) and resumable. Code, configurations and the trajectory construction are released with the paper.